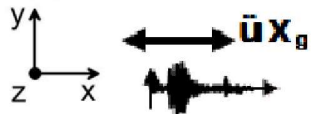
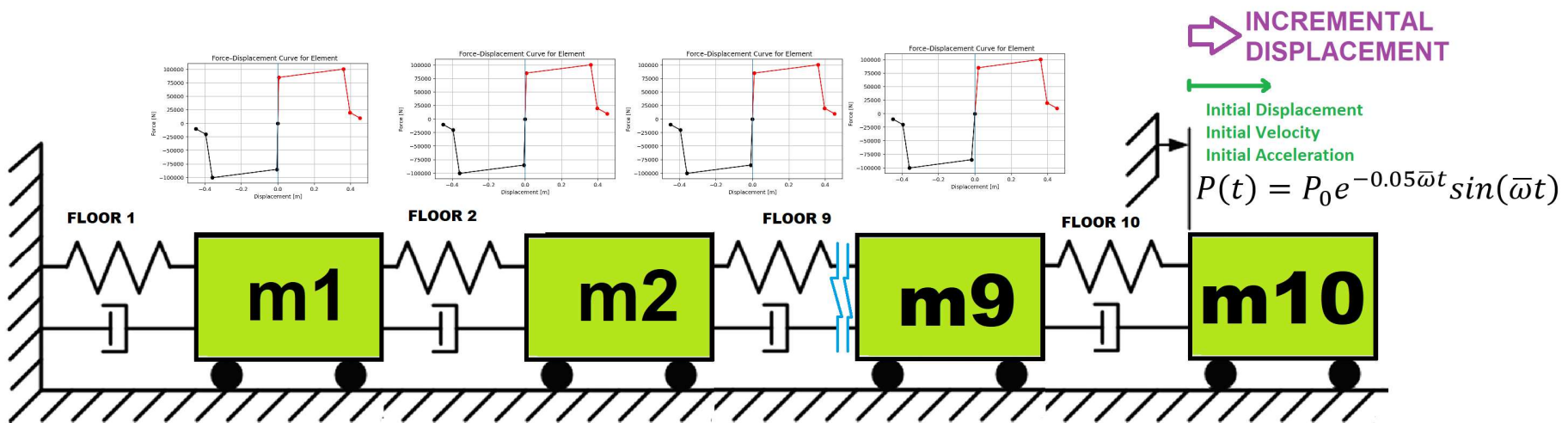


>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

SENSITIVITY ANALYSIS OF COLUMN DUCTILITY RATIO AND OPENSEES VIA 20 SEISMIC GROUND MOTIONS

THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)

Assume that a multi-degree-of-freedom structure is subjected to nonlinear dynamic analysis under twenty different ground motion records. To investigate the influence of column ductility on seismic performance, a sensitivity analysis is performed by systematically varying each column's ductility ratio from 20 to 50 in 21 steps. For each ductility level, a pushover analysis provides the capacity curve and equivalent SDOF parameters, a cyclic displacement analysis establishes a reference hysteretic energy, and twenty nonlinear time-history analyses are conducted using different ground motions. From these analyses, the Energy Dissipation Capacity Index (EDCI) and other response metrics such as maximum displacement, velocity, acceleration, damage index, over-strength factor, ductility ratio, and equivalent viscous damping ratio are computed. The median response over the twenty ground motions is then evaluated for each ductility level. Finally, the results are assessed through trend plots, 3D contour surfaces, correlation heatmaps, Random Forest regression, and ANOVA sensitivity analysis to identify the optimal column ductility ratio that maximizes energy dissipation capacity while maintaining structural safety.



$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

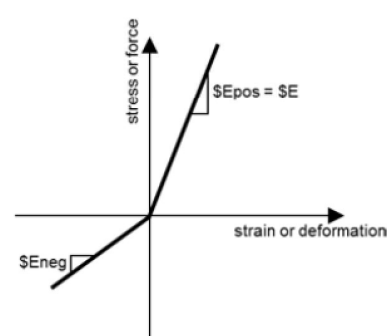
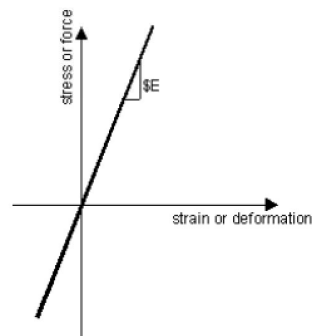
Δ_d = Lateral Displaement from Dynamic Analysis

Δ_y = Lateral Yield Displaement from Pushover Analysis

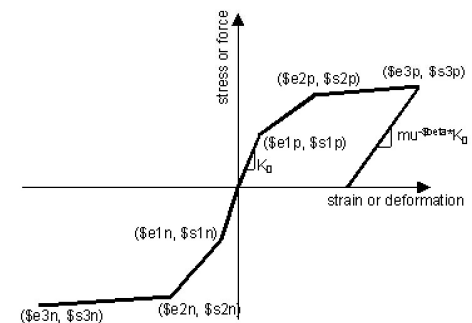
Δ_u = Lateral Ultimate Displaement from Pushover Analysis

$$P(t) = P_0 e^{-0.05\bar{\omega}t} \sin(\bar{\omega}t)$$

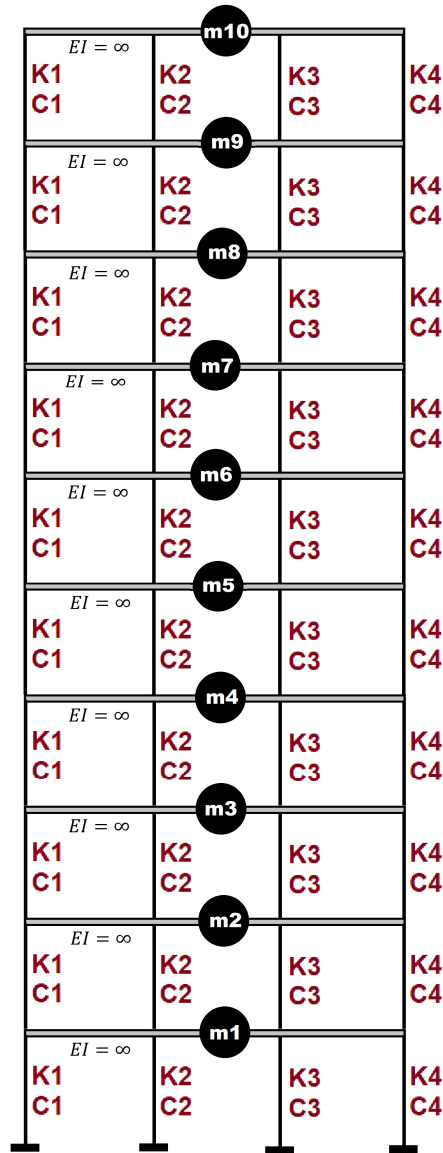
$$m\ddot{u} + c\dot{u} + ku = P(t)$$



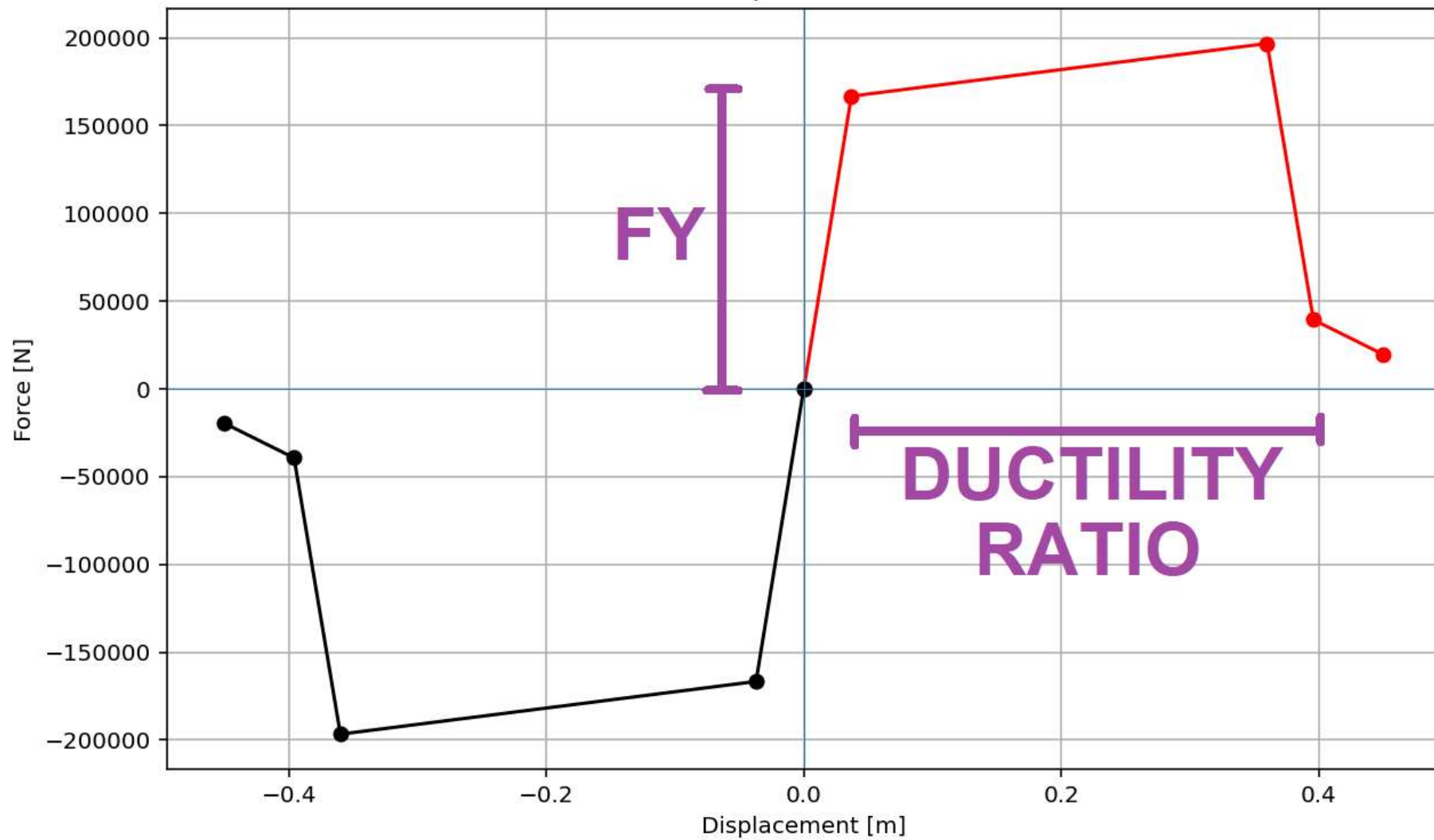
ELASTIC MATERIAL



INELASTIC MATERIAL



Force-Displacement Curve



C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...S_04_COLUMNS+\DUCT_SENSITIVITY_20_SEISMIC_MDOF.py

DUCT_SENSITIVITY_20_SEISMIC_MDOF.py X

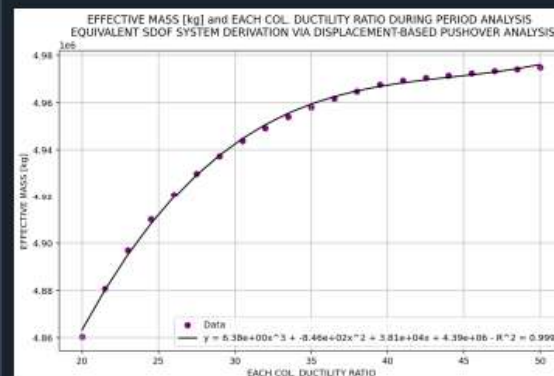
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1 #####
2 # >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
3 # SENSITIVITY ANALYSIS OF COLUMN DUCTILITY RATIO AND OPENSEES VIA 20 SEISMIC GROUND MOTIONS
4 # -----
5 # EQUIVALENT VISCOUS DAMPING RATIO:  $\xi_{eq} = 100 * E_d / (4 * \pi * E_s)$ 
6 # -----
7 # ENERGY DISSIPATION CAPACITY INDEX =  $100 * E_d(\text{earthquake}) / E_d(\text{cyclic displacement})$ 
8 # -----
9 #  $P(t) = P_0 \sin(\omega t)$ 
10 #  $P(t) = P_0 \exp(-0.05\omega t) \sin(\omega t)$ 
11 # -----
12 # ASSESSMENT OF DUCTILITY DAMAGE INDICES FOR STRUCTURAL ELEMENTS AND SYSTEMS AND EVALUATION
13 # OF ENERGY DISSIPATION CAPACITY INDEX
14 # -----
15 # THIS PYTHON SCRIPT IS WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
16 # EMAIL: salar.d.ghashghaei@gmail.com
17 #####
18 """
19 Assume that a multi-degree-of-freedom structure is subjected to nonlinear dynamic analysis under
20 twenty different ground motion records. To investigate the influence of column ductility on
21 seismic performance, a sensitivity analysis is performed by systematically varying each column's
22 ductility ratio from 20 to 50 in 21 steps. For each ductility level, a pushover analysis provides
23 the capacity curve and equivalent SDOF parameters, a cyclic displacement analysis establishes a
24 reference hysteretic energy, and twenty nonlinear time-history analyses are conducted using
25 different ground motions. From these analyses, the Energy Dissipation Capacity Index (EDCI)
26 and other response metrics such as maximum displacement, velocity, acceleration, damage index,
27 over-strength factor, ductility ratio, and equivalent viscous damping ratio are computed.
28 The median response over the twenty ground motions is then evaluated for each ductility level.
29 Finally, the results are assessed through trend plots, 3D contour surfaces, correlation heatmaps,
30 Random Forest regression, and ANOVA sensitivity analysis to identify the optimal column ductility
31 ratio that maximizes energy dissipation capacity while maintaining structural safety.
32
33 Nonlinear Seismic Performance Assessment of a MDOF

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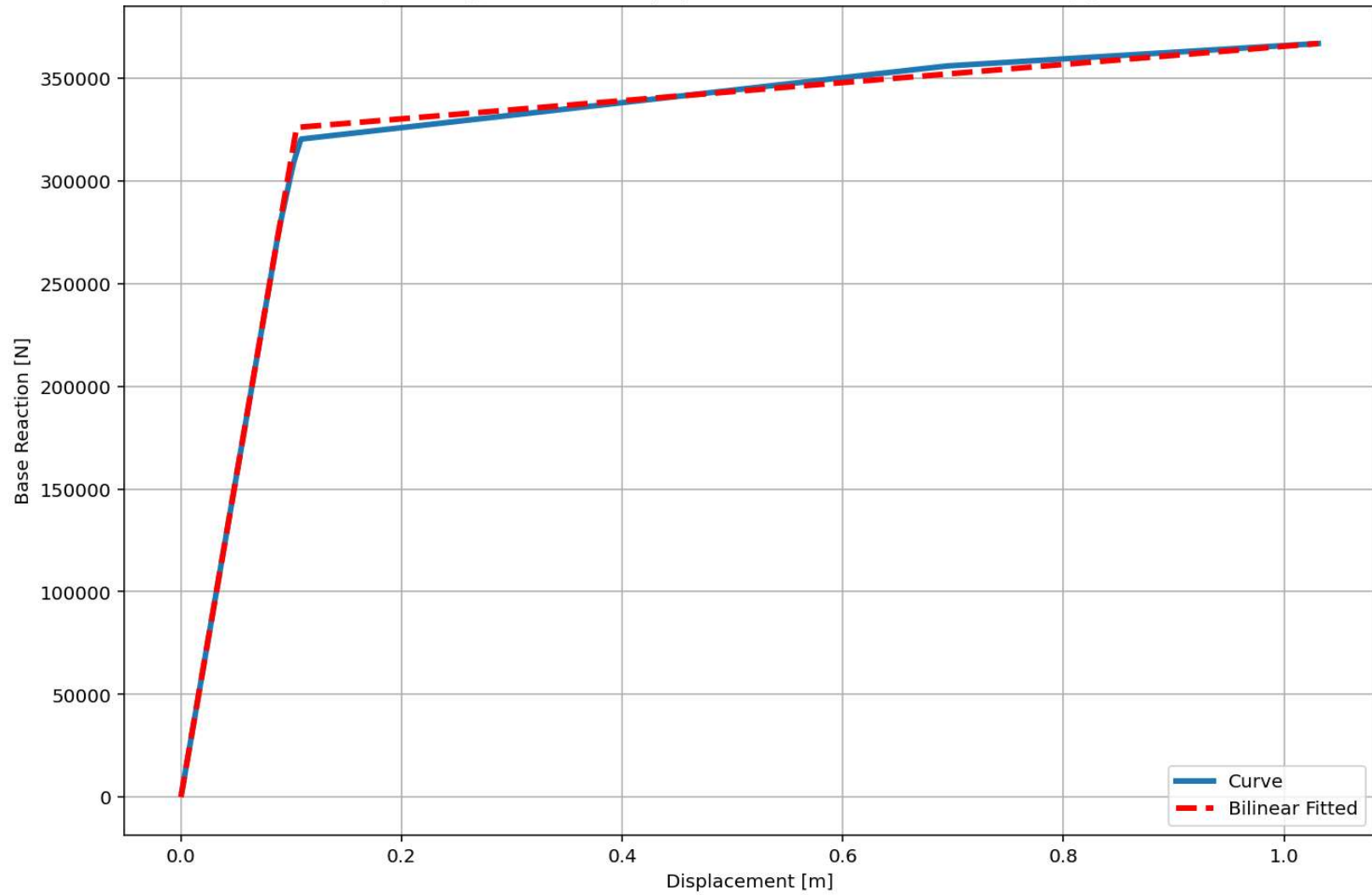
...A\OPTIMIZATION_20_SEISMIC_EDCI_MDOF_10_FLOORS_04_COLUMNS+

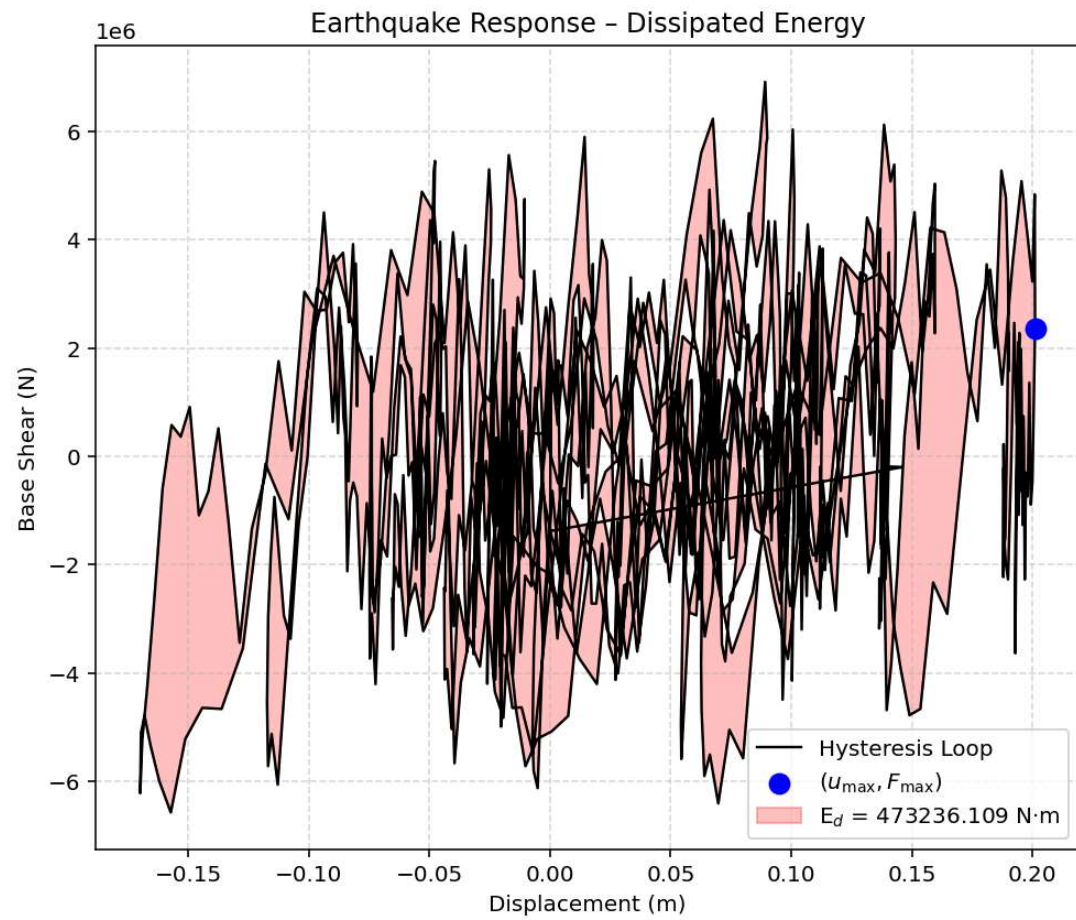
29 %

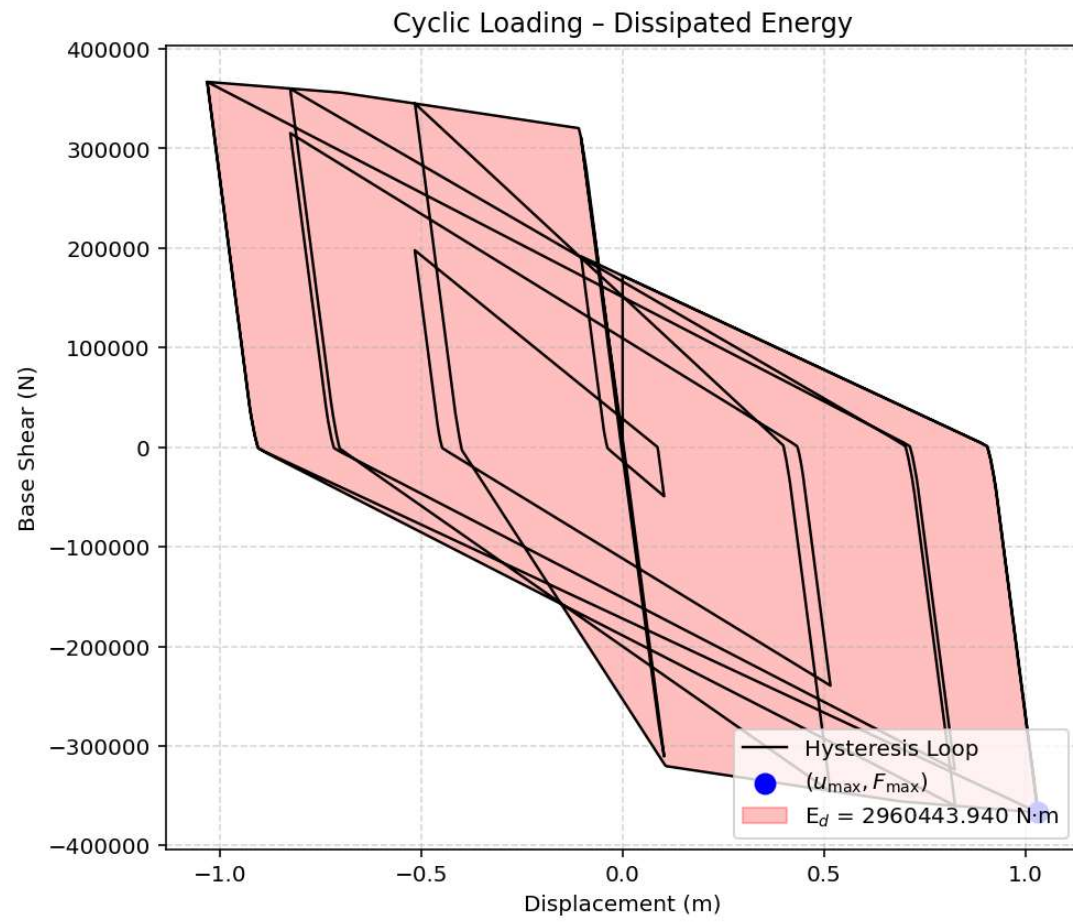


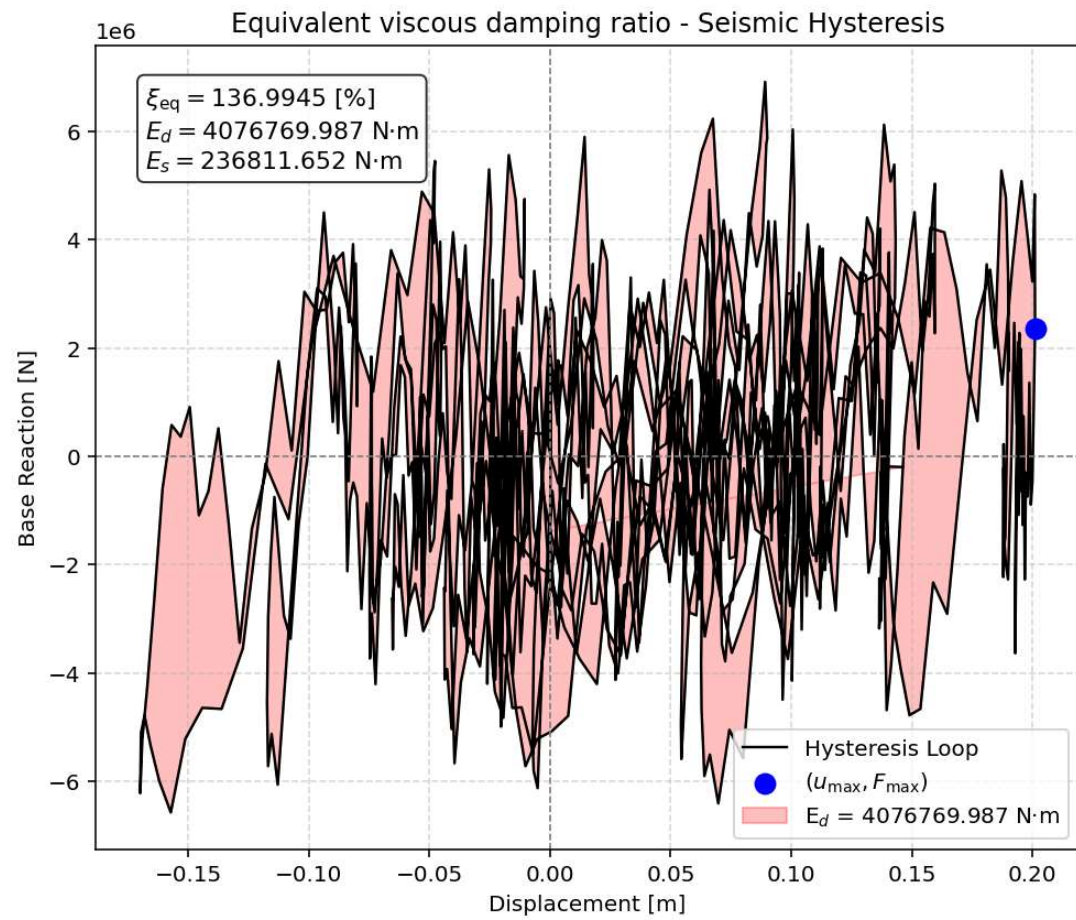
IPython Console Plots Variable Explorer Help Debugger History Files

BaseShear-Displacement Analysis - Ductility Ratio: 9.8321 - Over Strength Factor: 1.1254
Ductility Damage Index: 10.39 [%] - Structural Behavior Coefficient (R): 11.0646

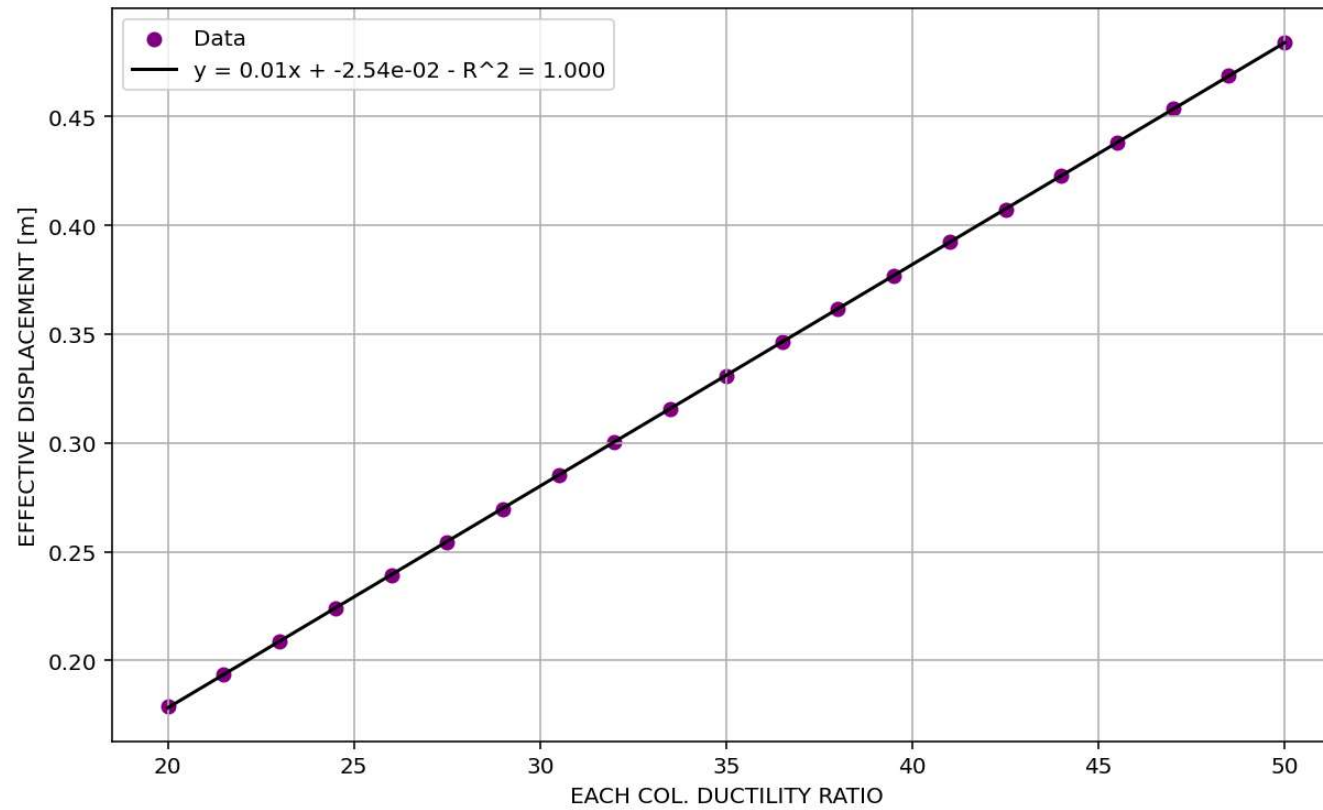




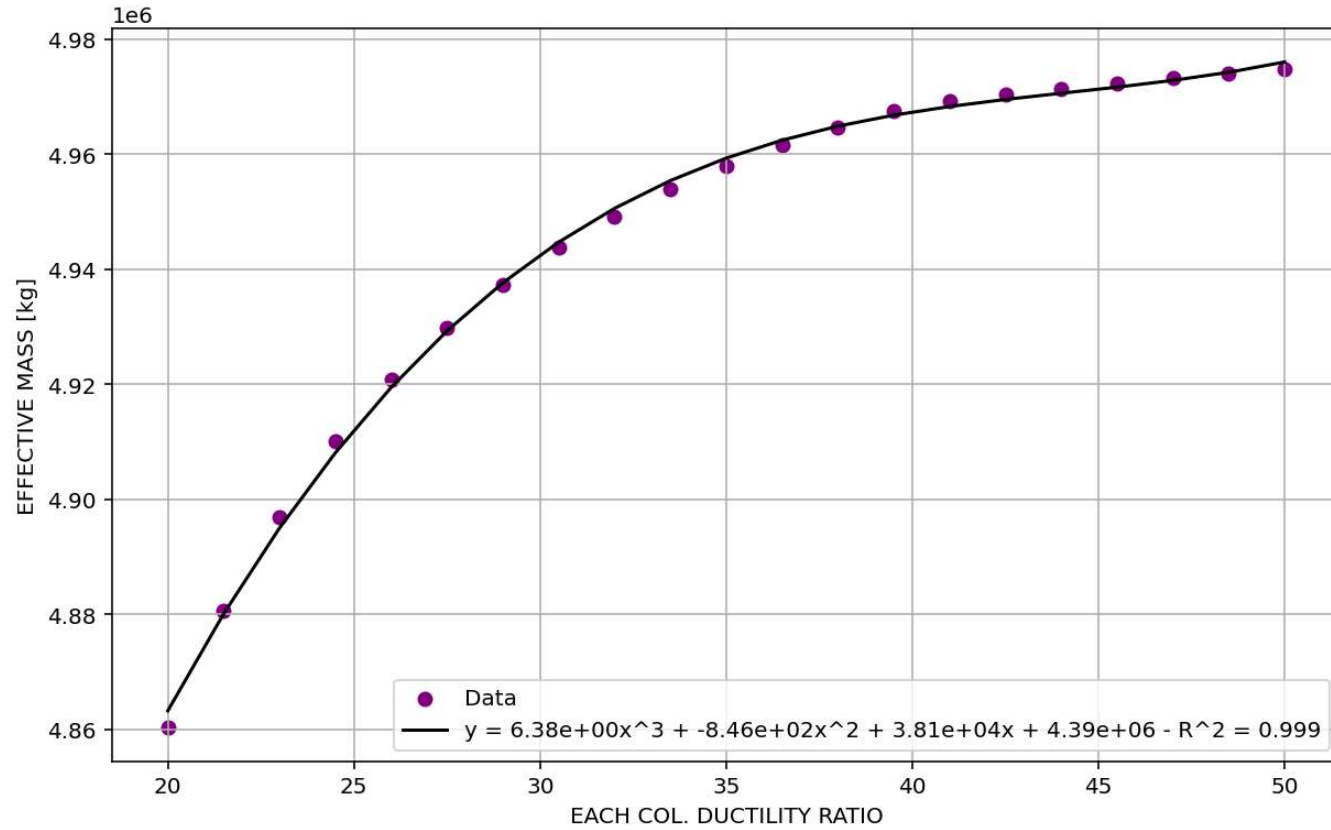




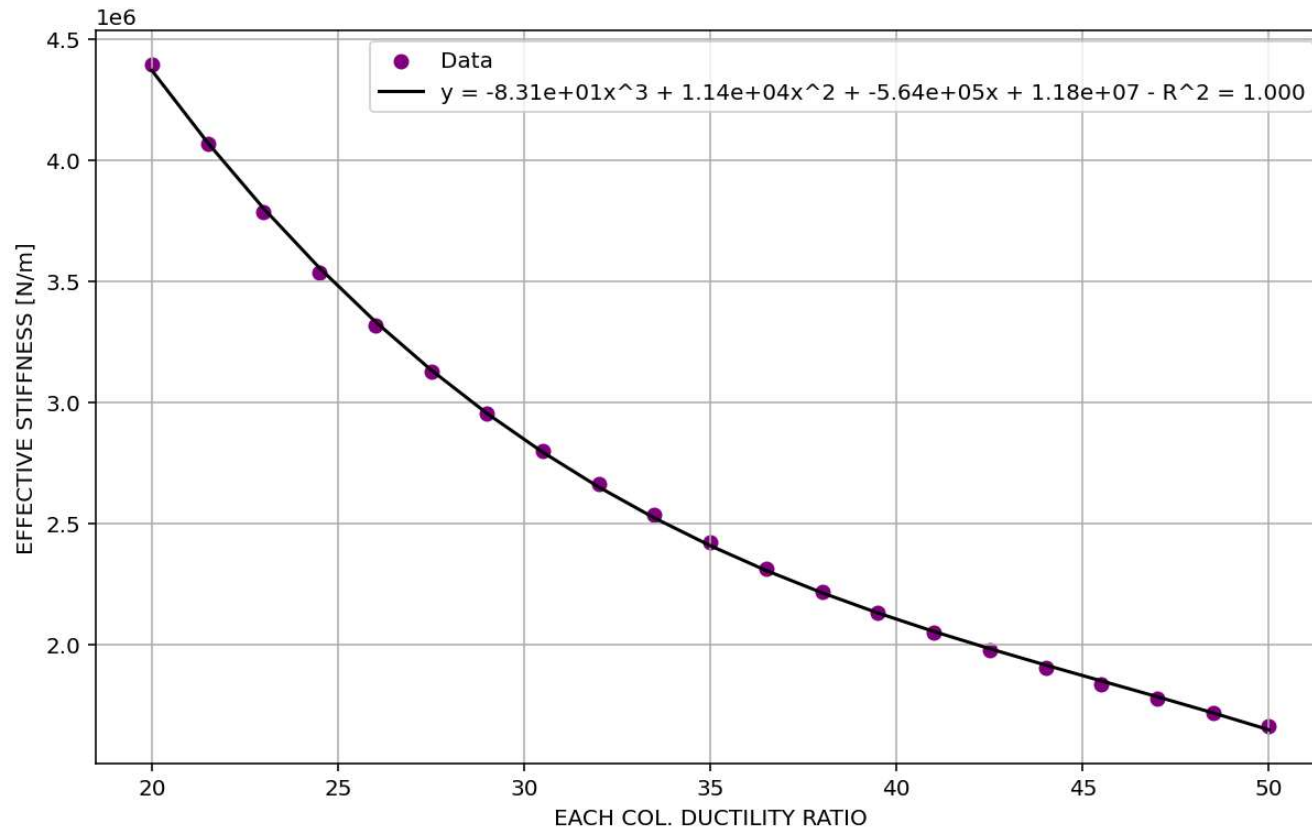
EFFECTIVE DISPLACEMENT [m] and EACH COL. DUCTILITY RATIO DURING PERIOD ANALYSIS
EQUIVALENT SDOF SYSTEM DERIVATION VIA DISPLACEMENT-BASED PUSHOVER ANALYSIS



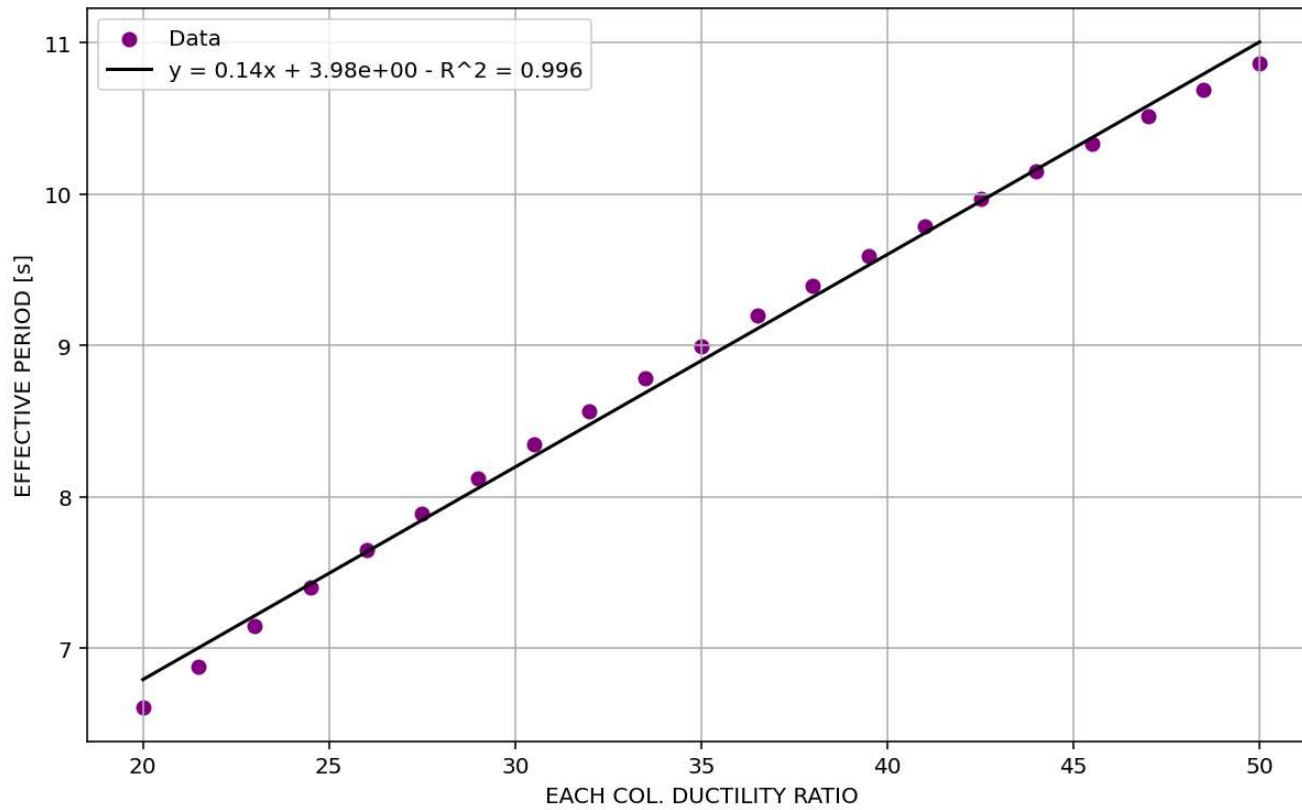
EFFECTIVE MASS [kg] and EACH COL. DUCTILITY RATIO DURING PERIOD ANALYSIS
EQUIVALENT SDOF SYSTEM DERIVATION VIA DISPLACEMENT-BASED PUSHOVER ANALYSIS



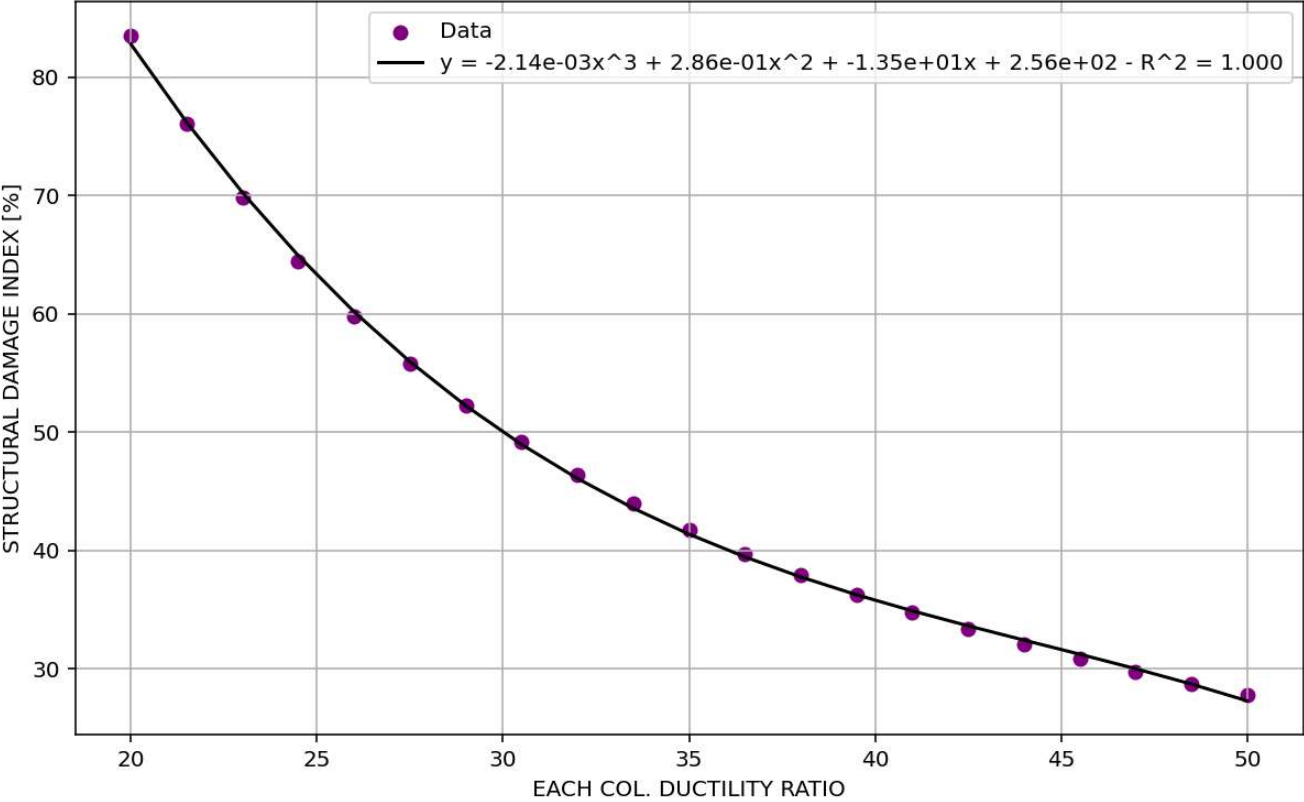
EFFECTIVE STIFFNESS [N/m] and EACH COL. DUCTILITY RATIO DURING PERIOD ANALYSIS
EQUIVALENT SDOF SYSTEM DERIVATION VIA DISPLACEMENT-BASED PUSHOVER ANALYSIS



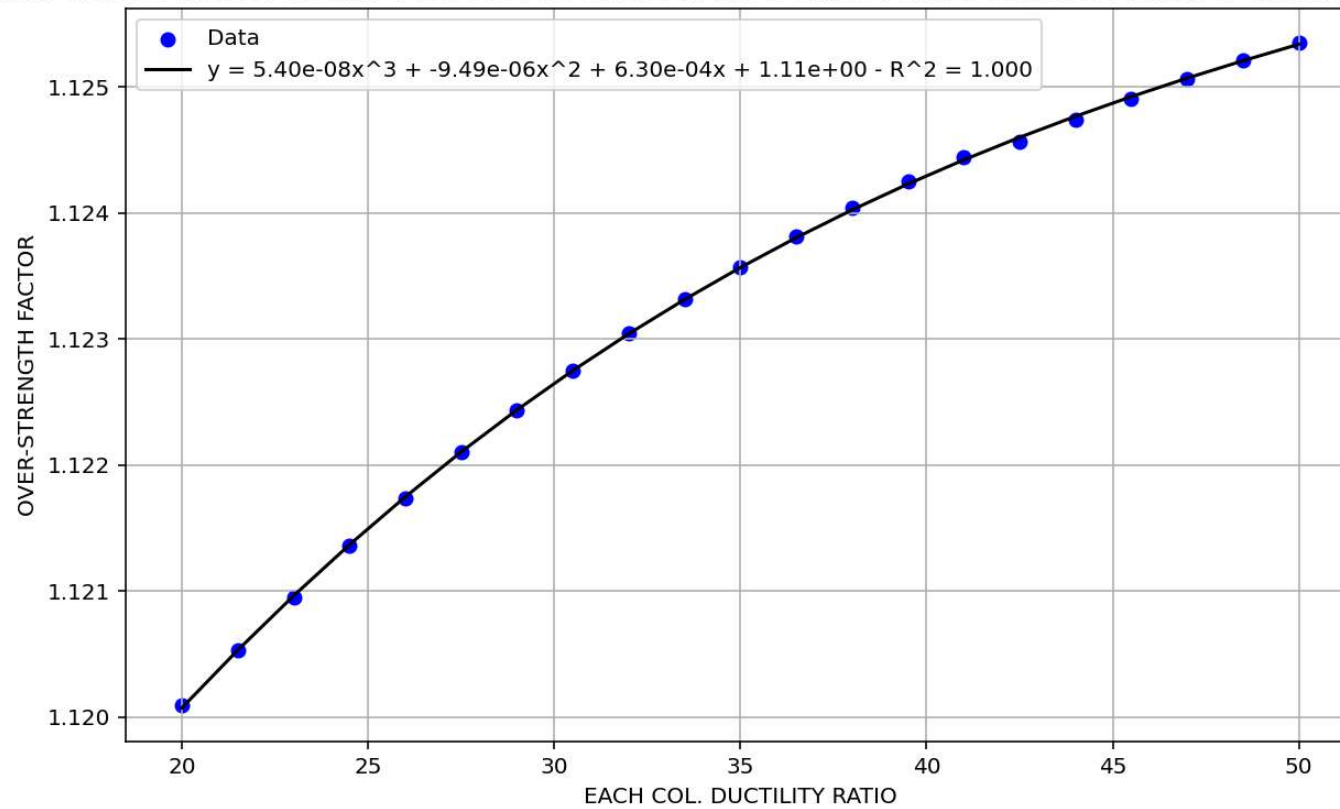
EFFECTIVE PERIOD [s] and EACH COL. DUCTILITY RATIO DURING PERIOD ANALYSIS
EQUIVALENT SDOF SYSTEM DERIVATION VIA DISPLACEMENT-BASED PUSHOVER ANALYSIS



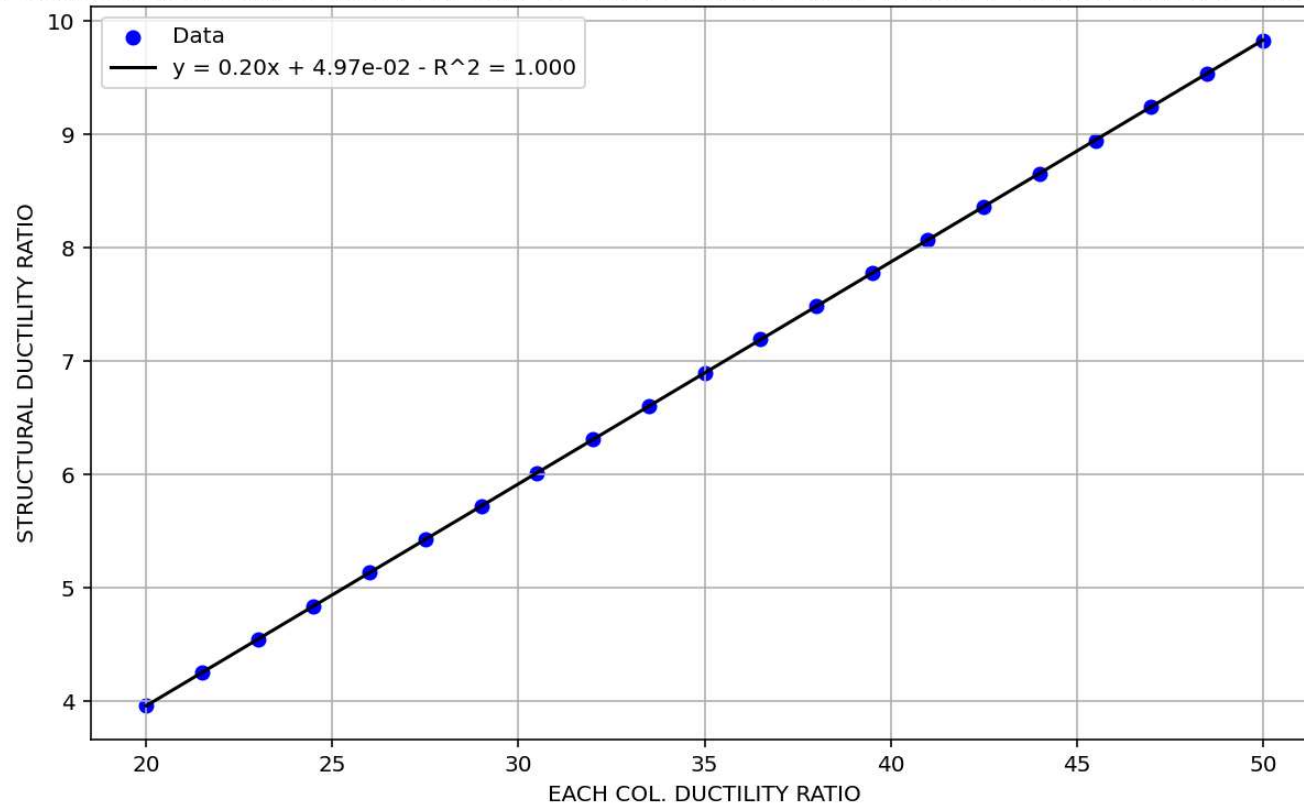
STRUCTURAL DAMAGE INDEX [%] and EACH COL. DUCTILITY RATIO DURING PERIOD ANALYSIS WITH 20 GROUND MOTIONS



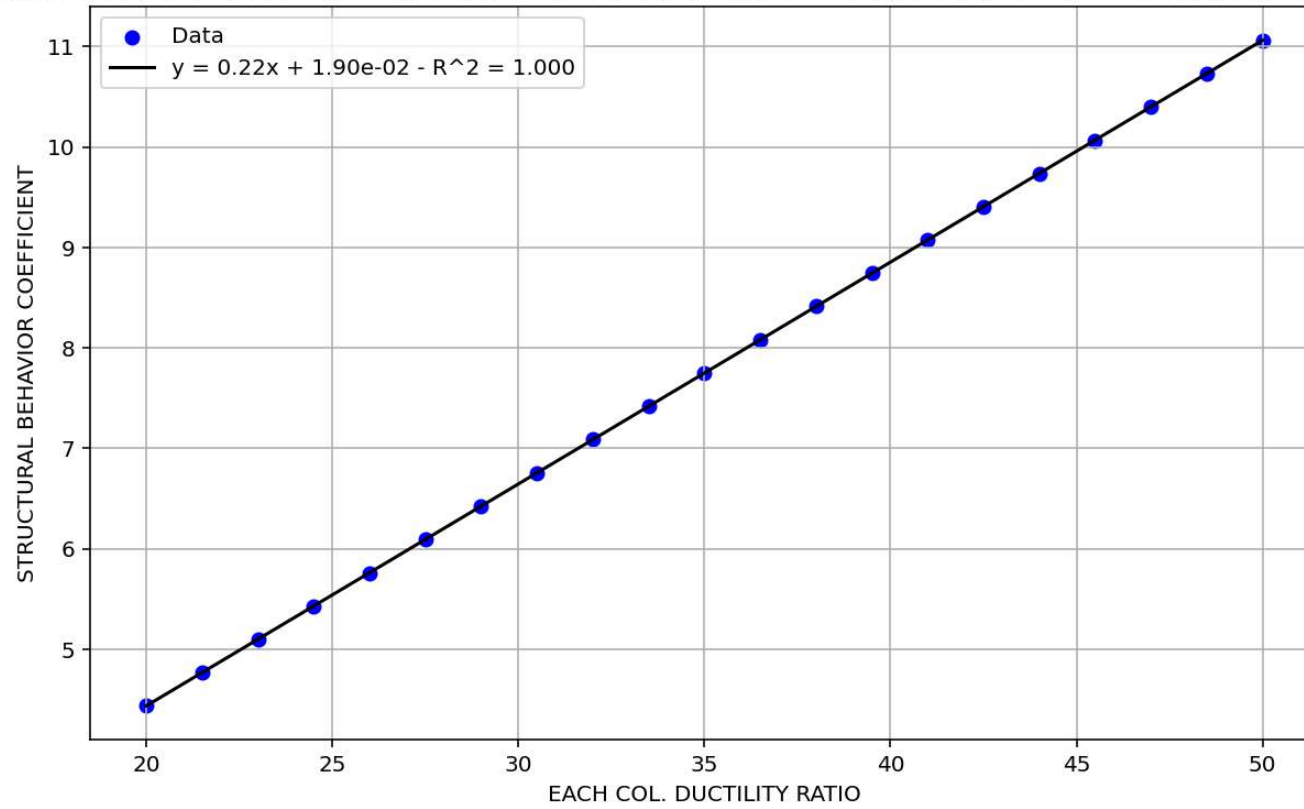
OVER-STRENGTH FACTOR and EACH COL. DUCTILITY RATIO DURING PERIOD ANALYSIS WITH 20 GROUND MOTIONS



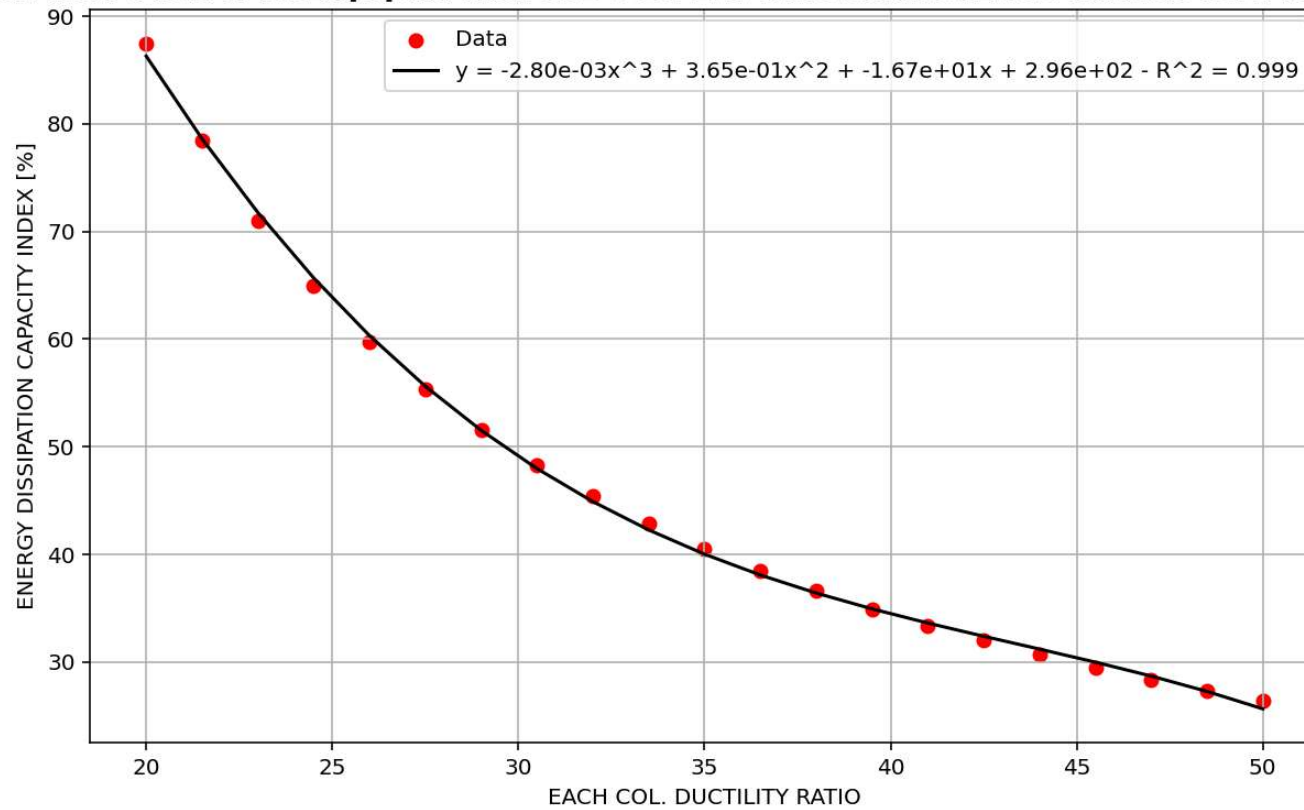
STRUCTURAL DUCTILITY RATIO and EACH COL. DUCTILITY RATIO DURING PERIOD ANALYSIS WITH 20 GROUND MOTIONS



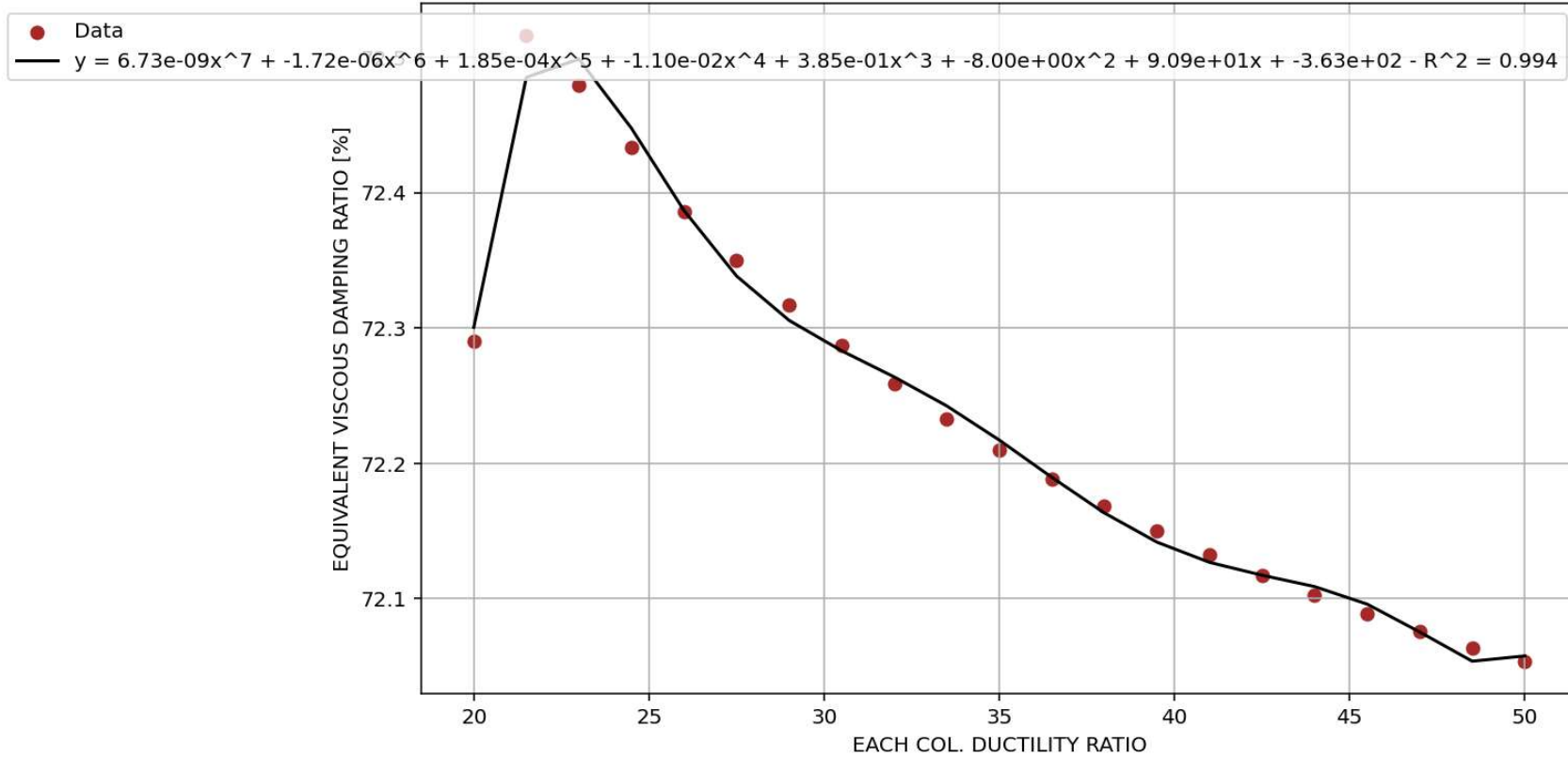
STRUCTURAL BEHAVIOR COEFFICIENT and EACH COL. DUCTILITY RATIO DURING PERIOD ANALYSIS WITH 20 GROUND MOTIONS



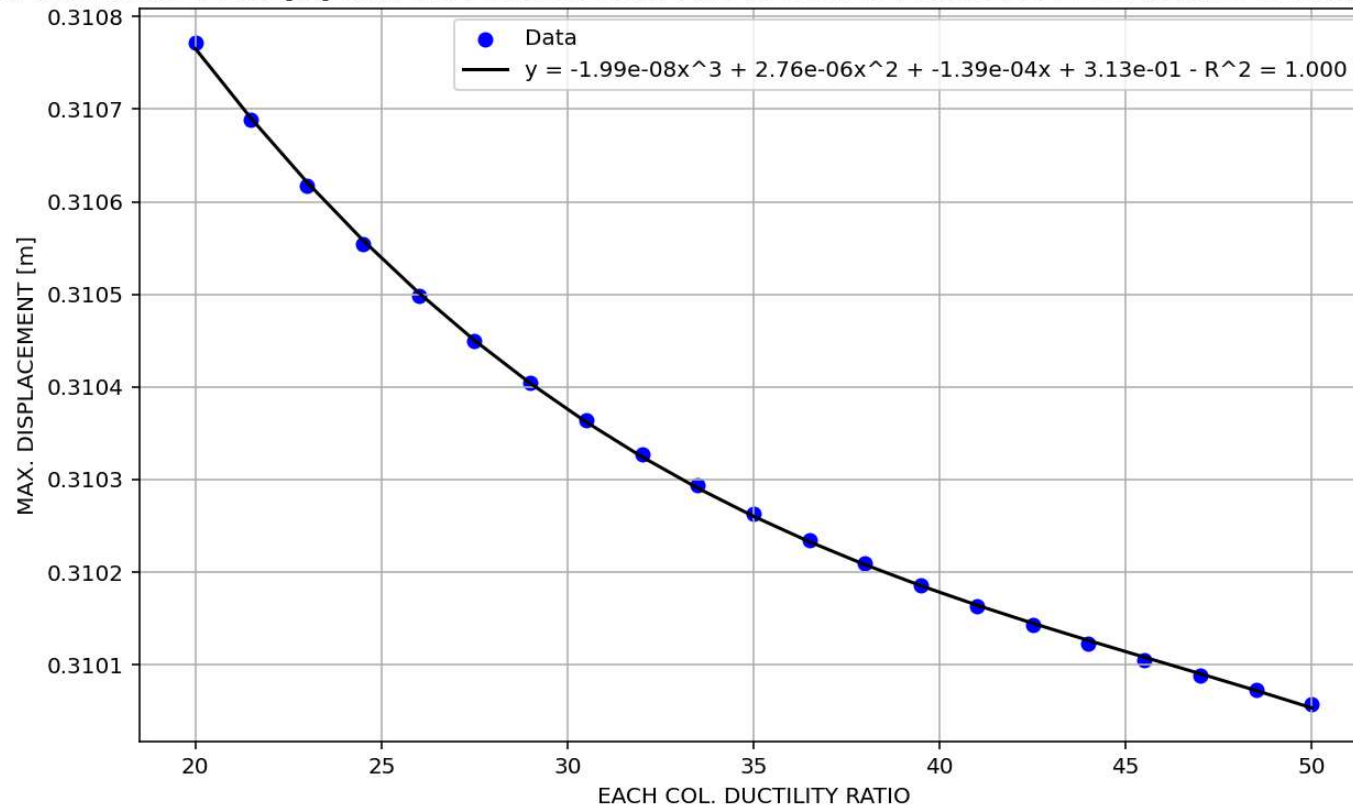
ENERGY DISSIPATION CAPACITY INDEX [%] and EACH COL. DUCTILITY RATIO DURING PERIOD ANALYSIS WITH 20 GROUND MOTIONS



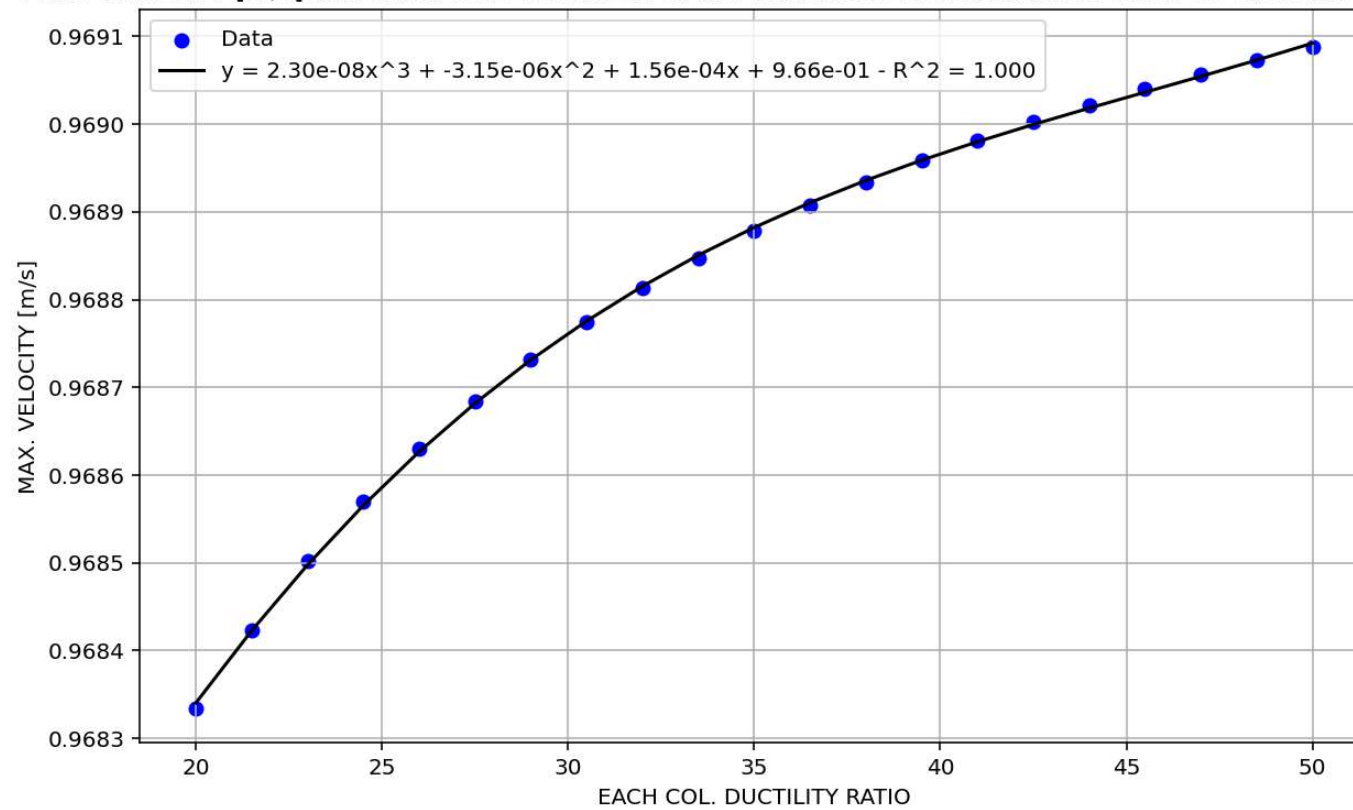
EQUIVALENT VISCOUS DAMPING RATIO [%] and EACH COL. DUCTILITY RATIO DURING PERIOD ANALYSIS WITH 20 GROUND MOTIONS



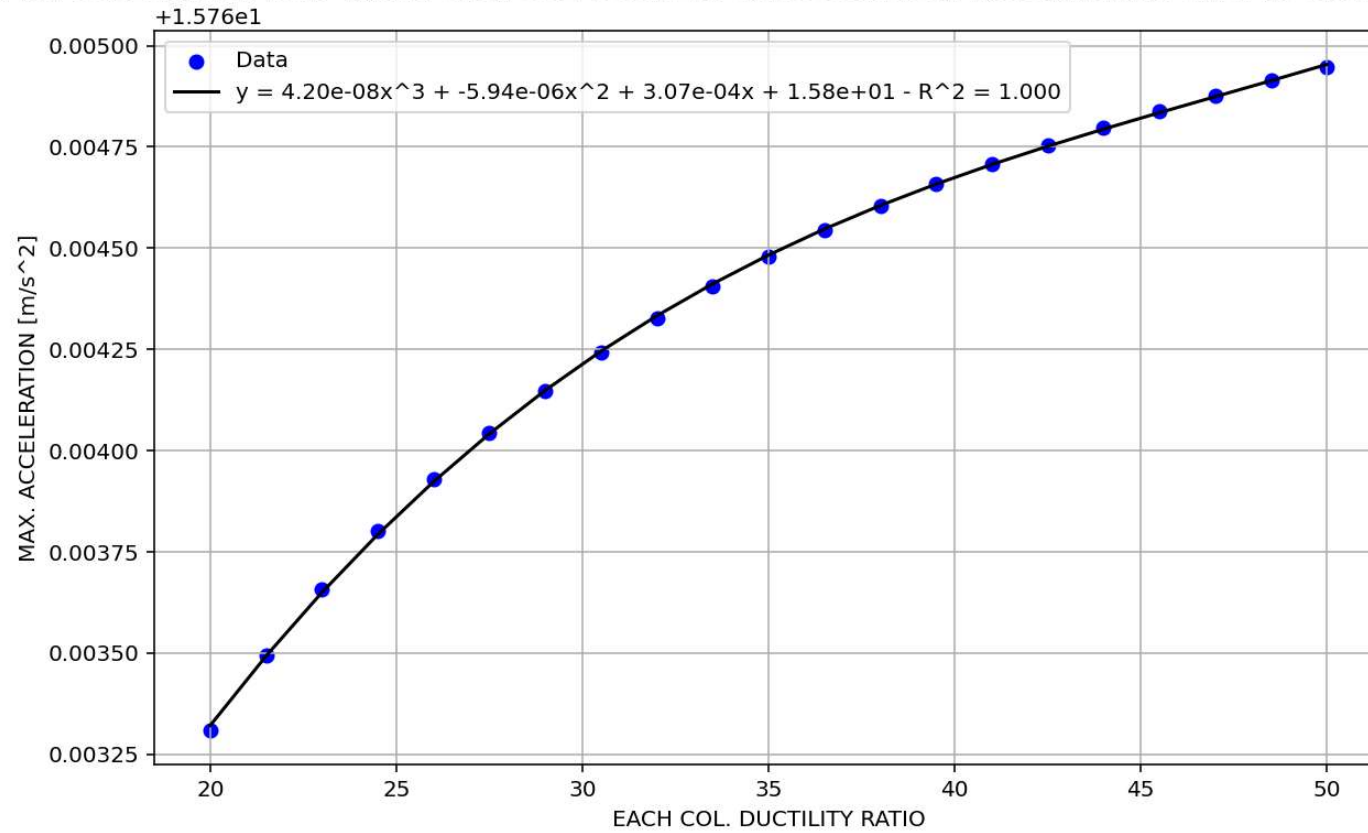
MAX. DISPLACEMENT [m] and EACH COL. DUCTILITY RATIO DURING PERIOD ANALYSIS WITH 20 GROUND MOTIONS



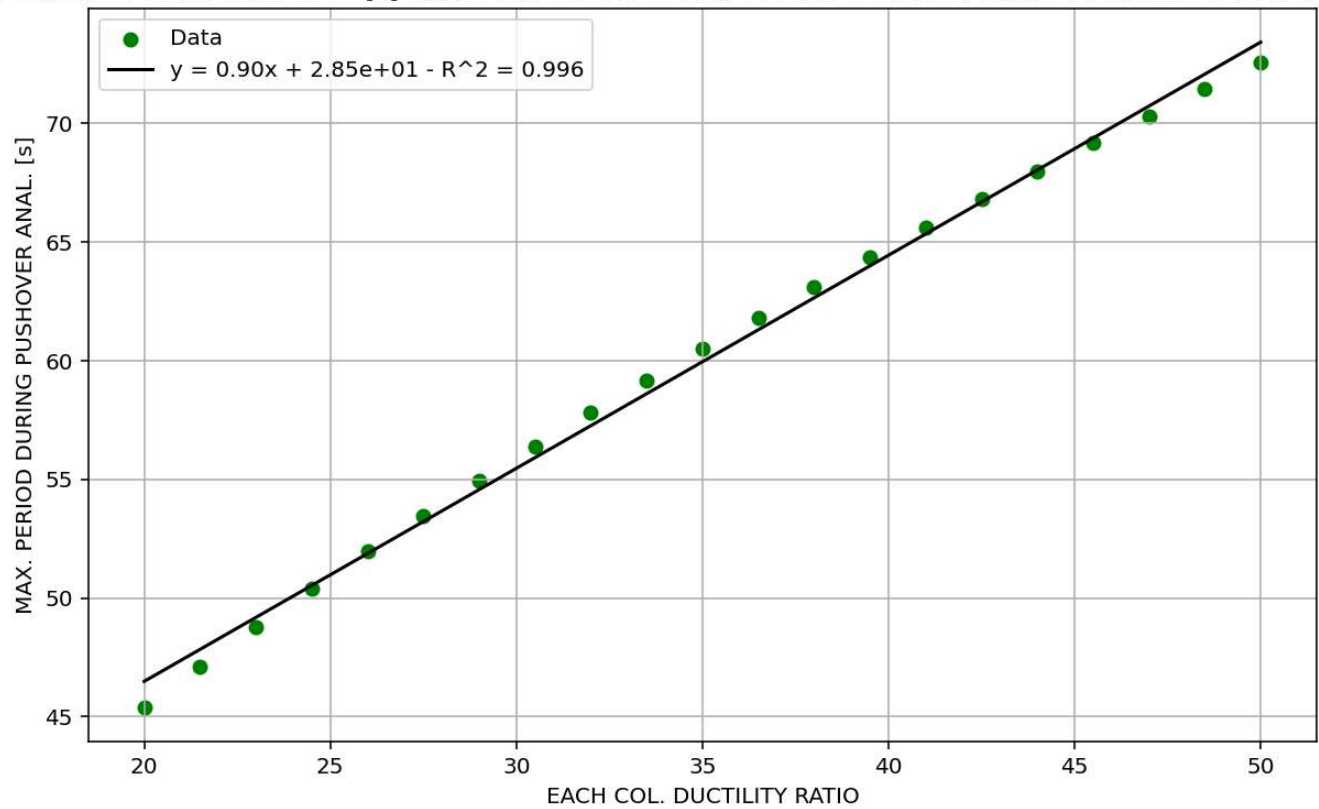
MAX. VELOCITY [m/s] and EACH COL. DUCTILITY RATIO DURING PERIOD ANALYSIS WITH 20 GROUND MOTIONS



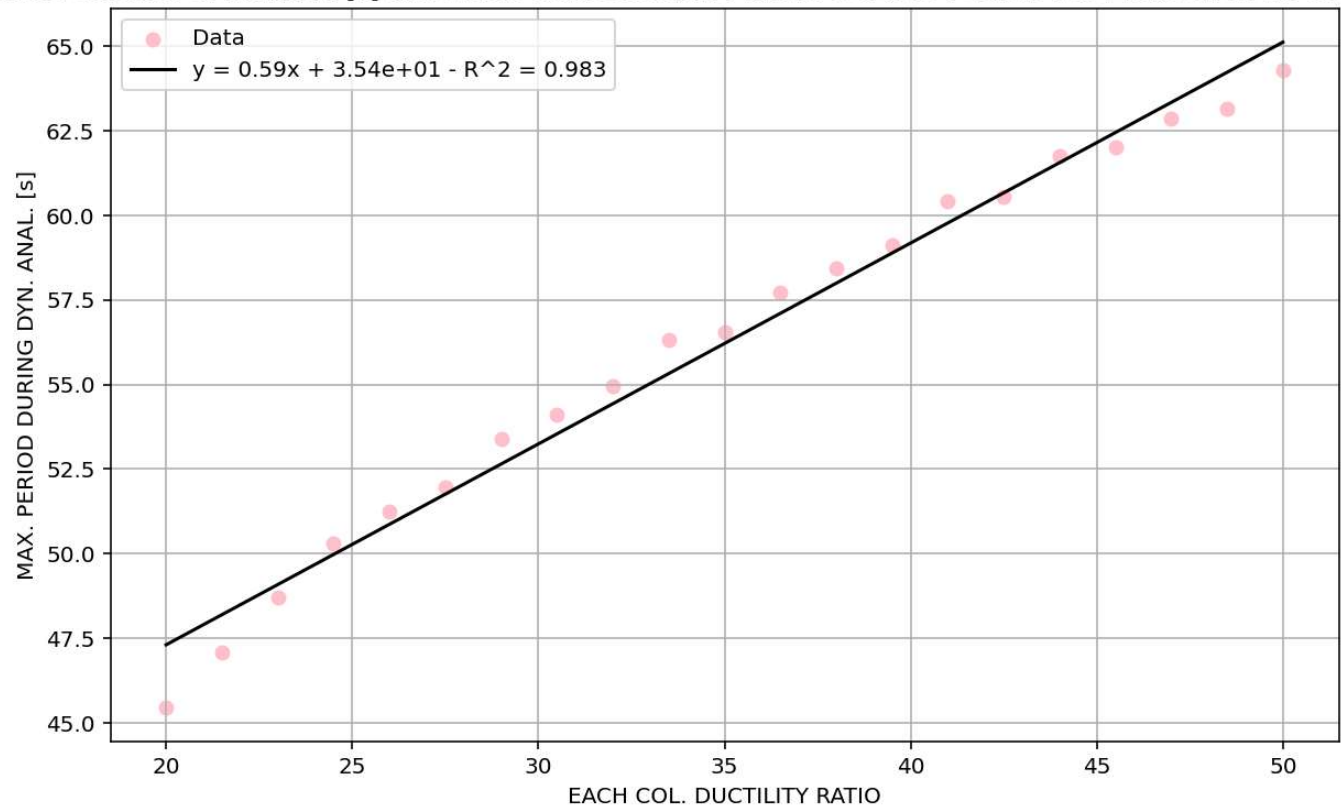
MAX. ACCELERATION [m/s²] and EACH COL. DUCTILITY RATIO DURING PERIOD ANALYSIS WITH 20 GROUND MOTIONS



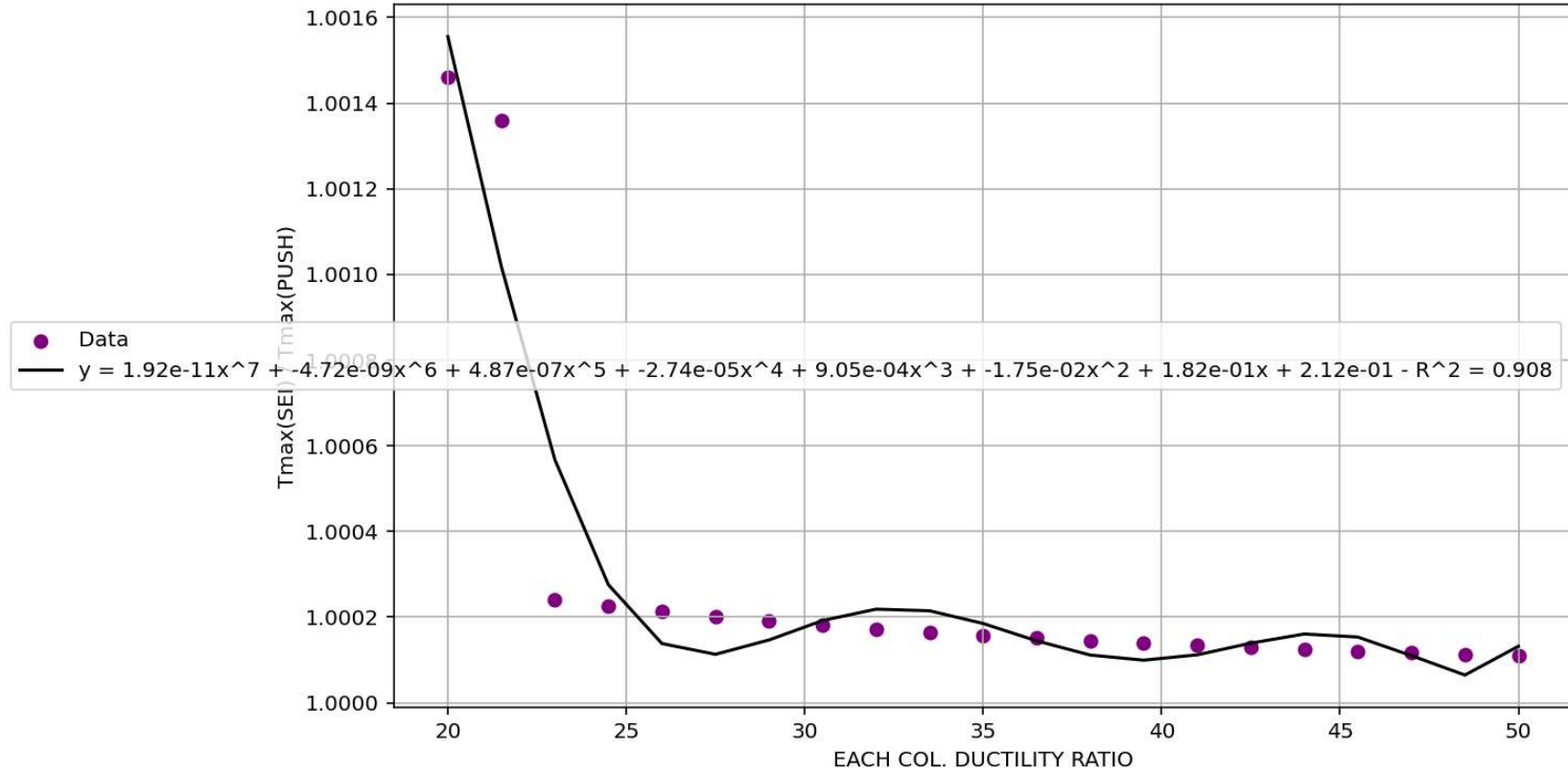
MAX. PERIOD DURING PUSHOVER ANAL. [s] and EACH COL. DUCTILITY RATIO DURING PERIOD ANALYSIS WITH 20 GROUND MOTIONS



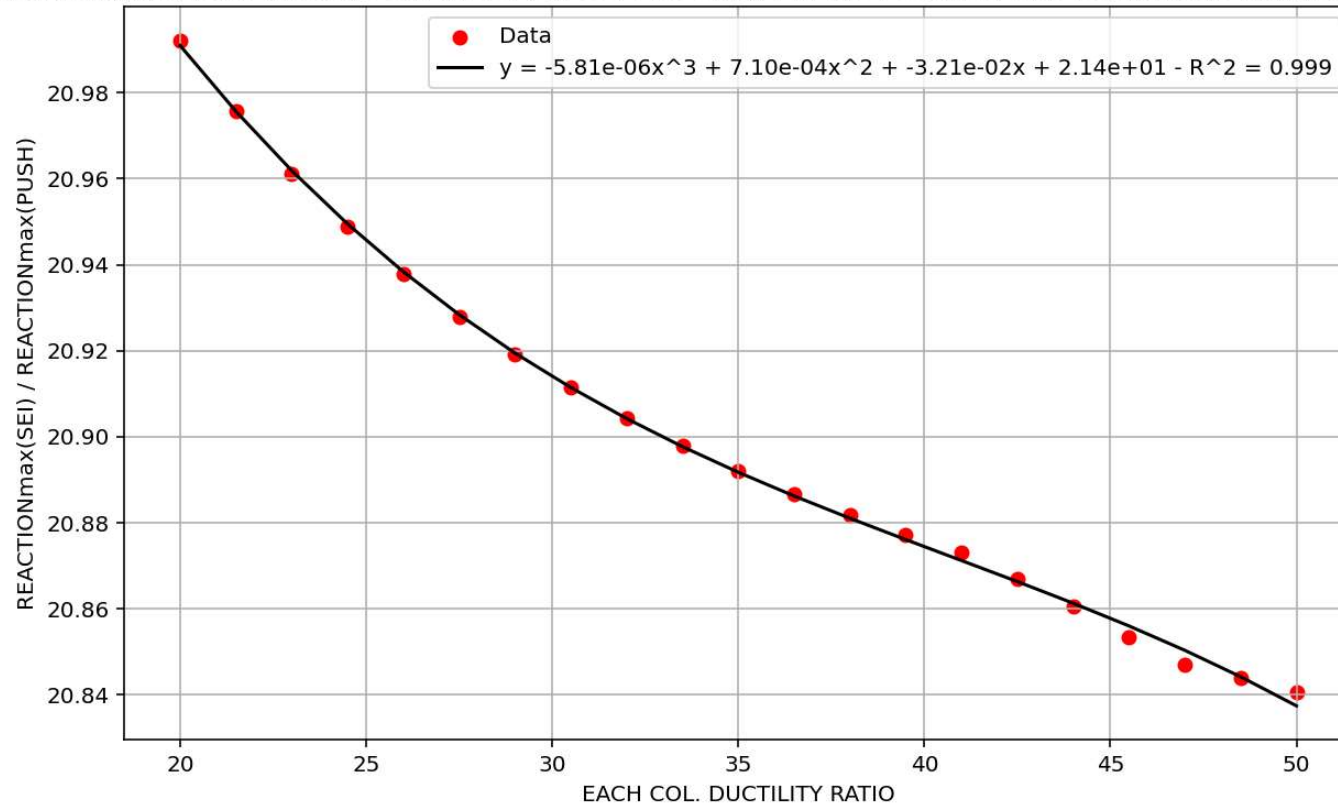
MAX. PERIOD DURING DYN. ANAL. [s] and EACH COL. DUCTILITY RATIO DURING PERIOD ANALYSIS WITH 20 GROUND MOTIONS



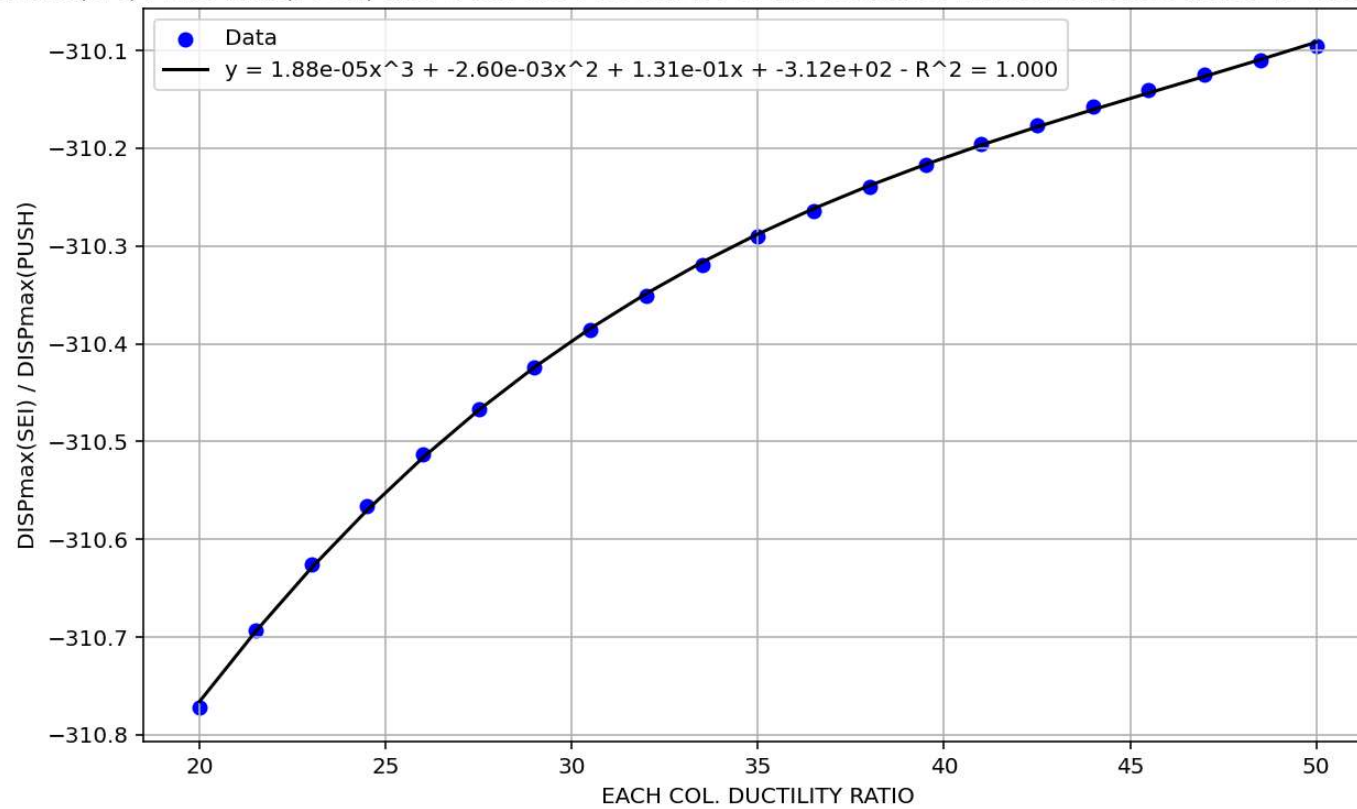
Tmax(SEI) / Tmax(PUSH) and EACH COL. DUCTILITY RATIO DURING PERIOD ANALYSIS WITH 20 GROUND MOTIONS



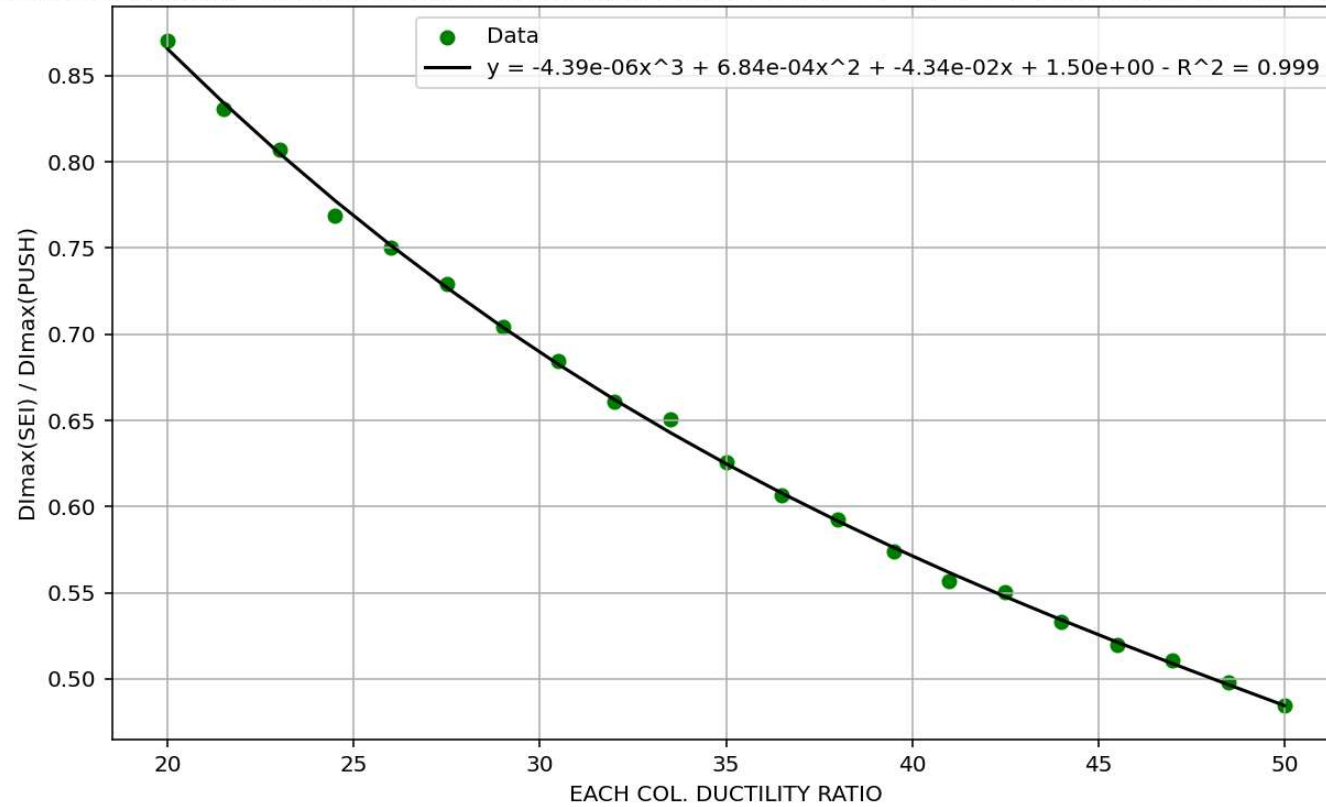
REACTIONmax(SEI) / REACTIONmax(PUSH) and EACH COL. DUCTILITY RATIO DURING PERIOD ANALYSIS WITH 20 GROUND MOTIONS



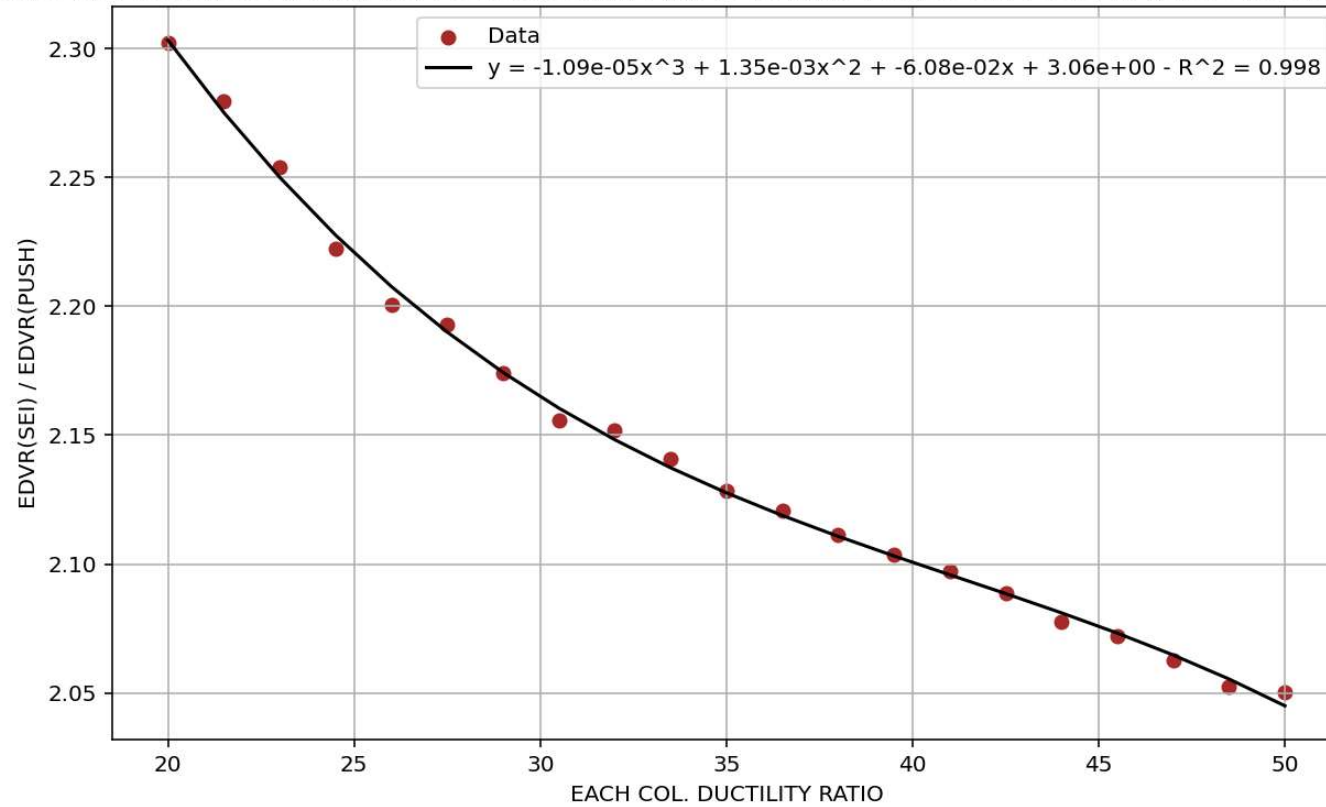
DISPmax(SEI) / DISPmax(PUSH) and EACH COL. DUCTILITY RATIO DURING PERIOD ANALYSIS WITH 20 GROUND MOTIONS



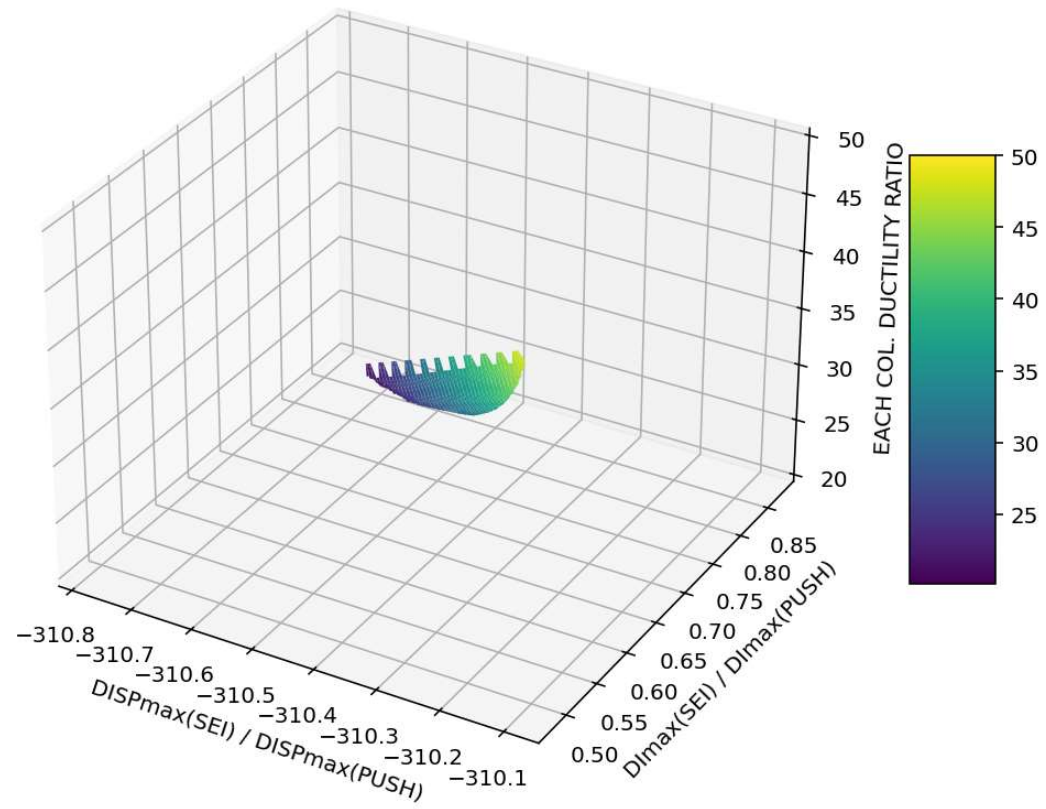
Dlmax(SEI) / Dlmax(PUSH) and EACH COL. DUCTILITY RATIO DURING PERIOD ANALYSIS WITH 20 GROUND MOTIONS

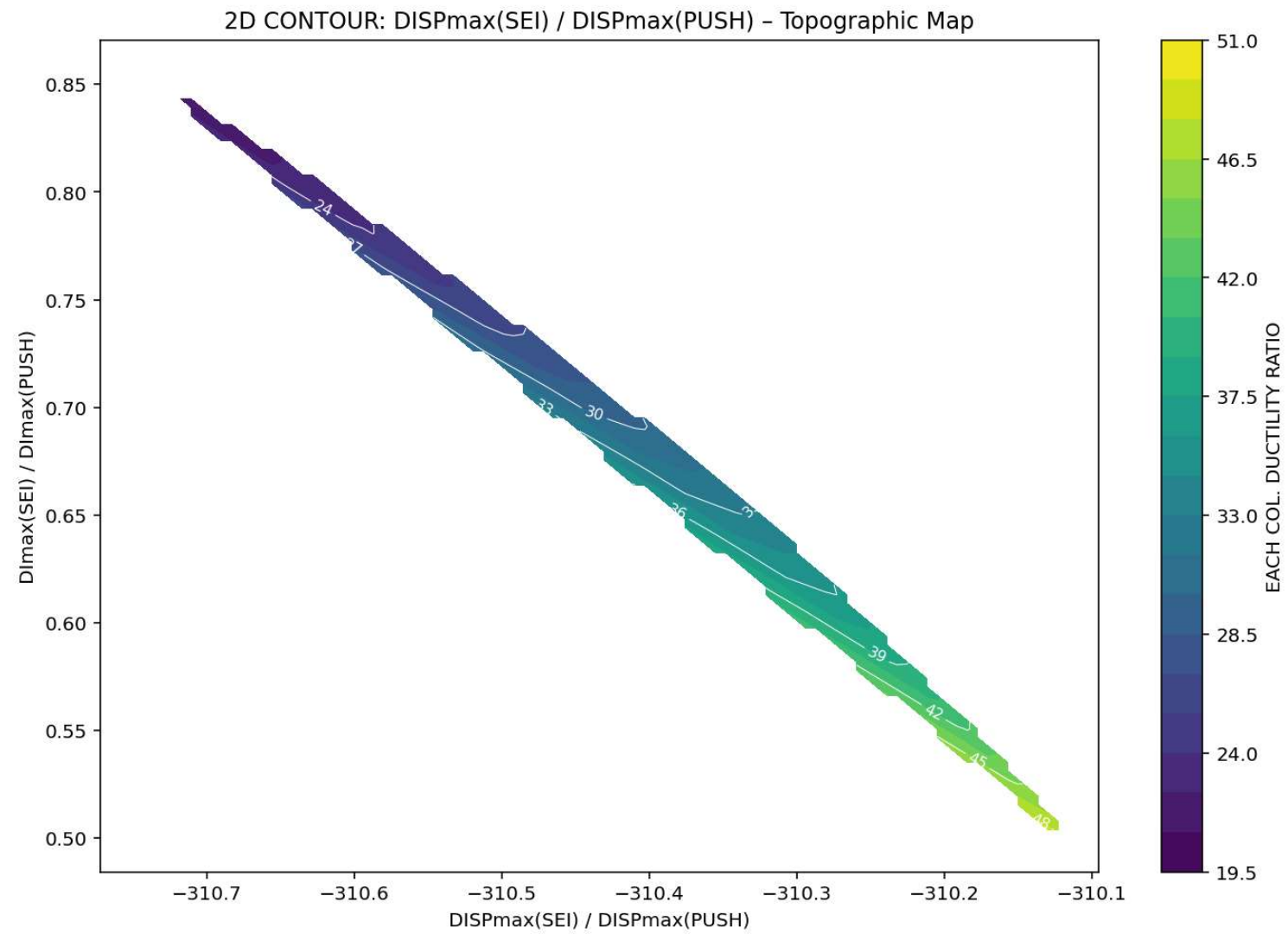


EDVR(SEI) / EDVR(PUSH) and EACH COL. DUCTILITY RATIO DURING PERIOD ANALYSIS WITH 20 GROUND MOTIONS

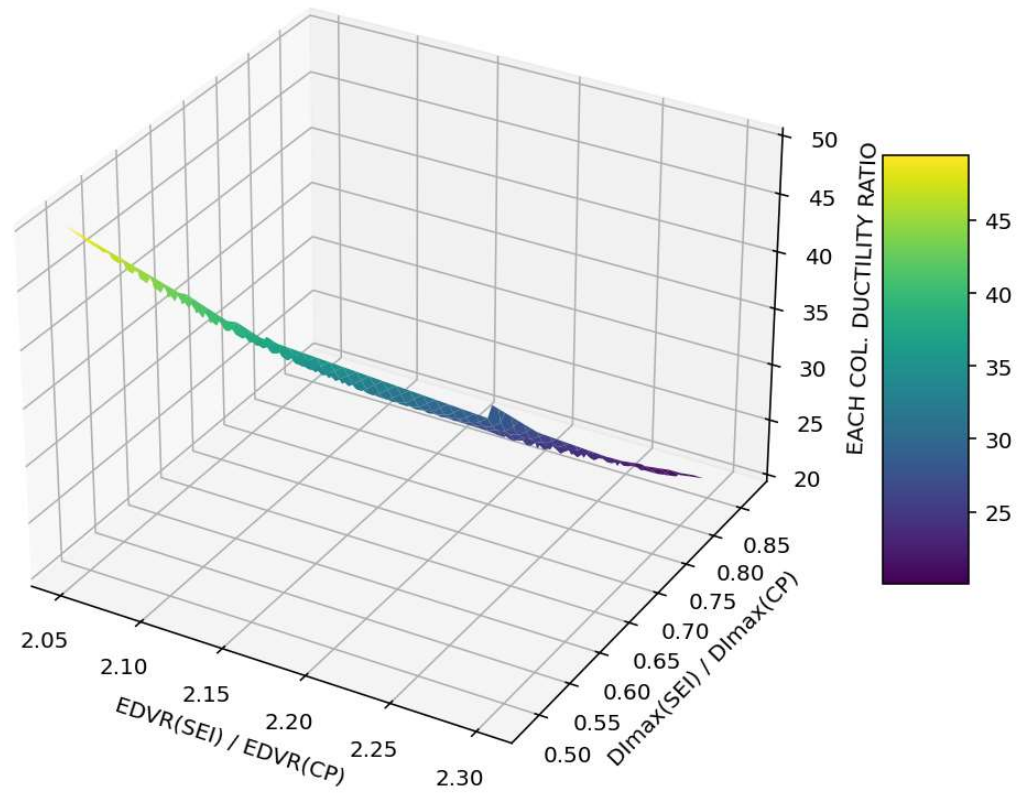


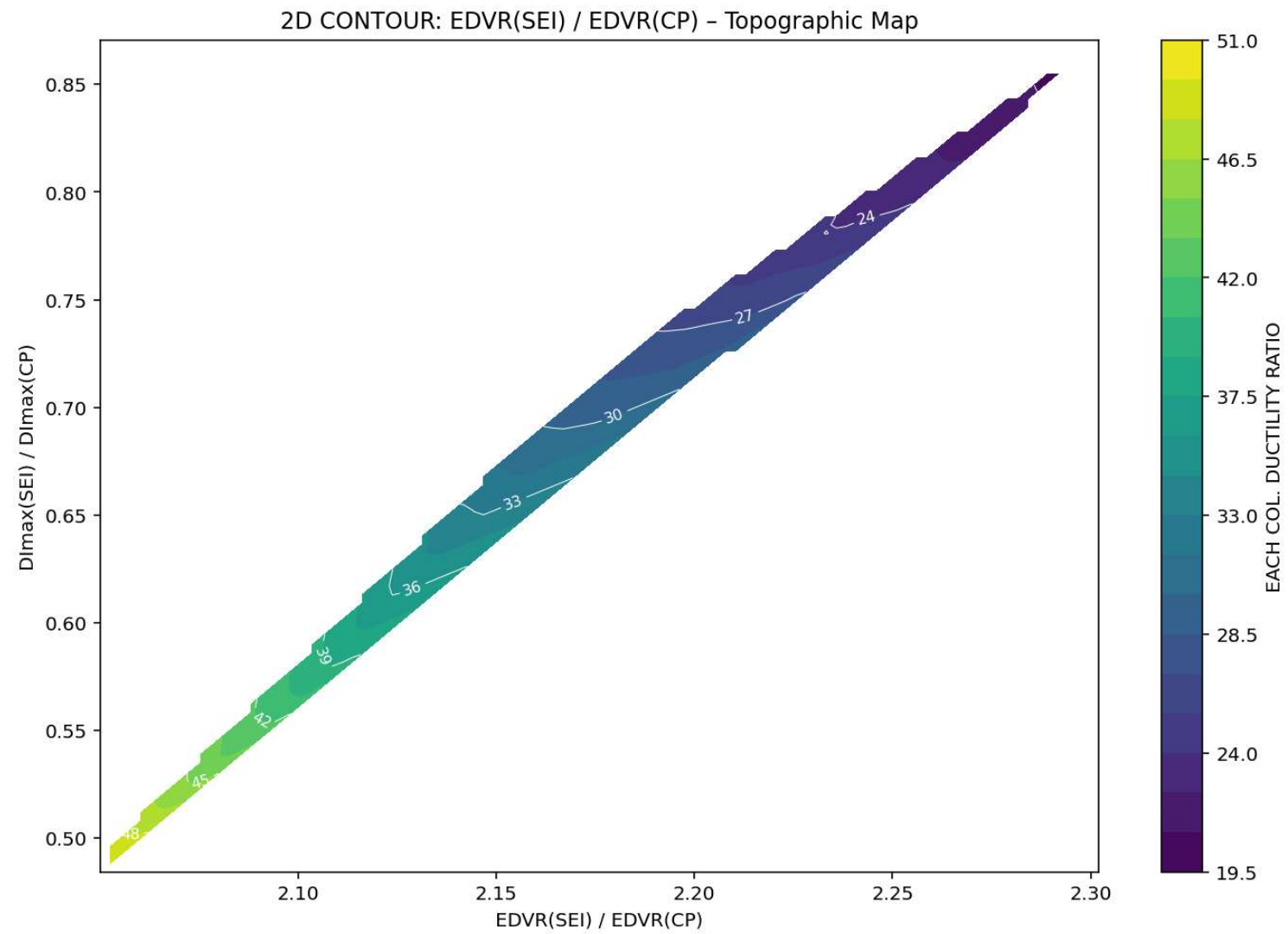
3D CONTOUR: $\text{DISPmax(SEI)} / \text{DISPmax(PUSH)}$ - $\text{Dlmax(SEI)} / \text{Dlmax(PUSH)}$ - EACH COL. DUCTILITY RATIO



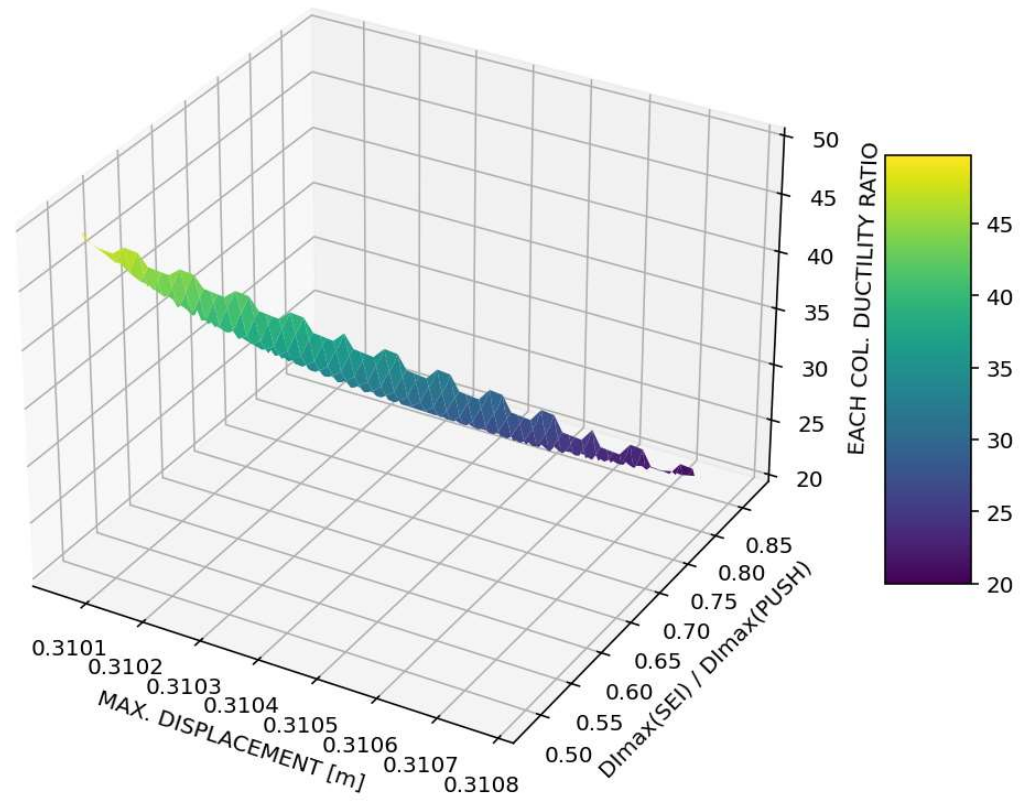


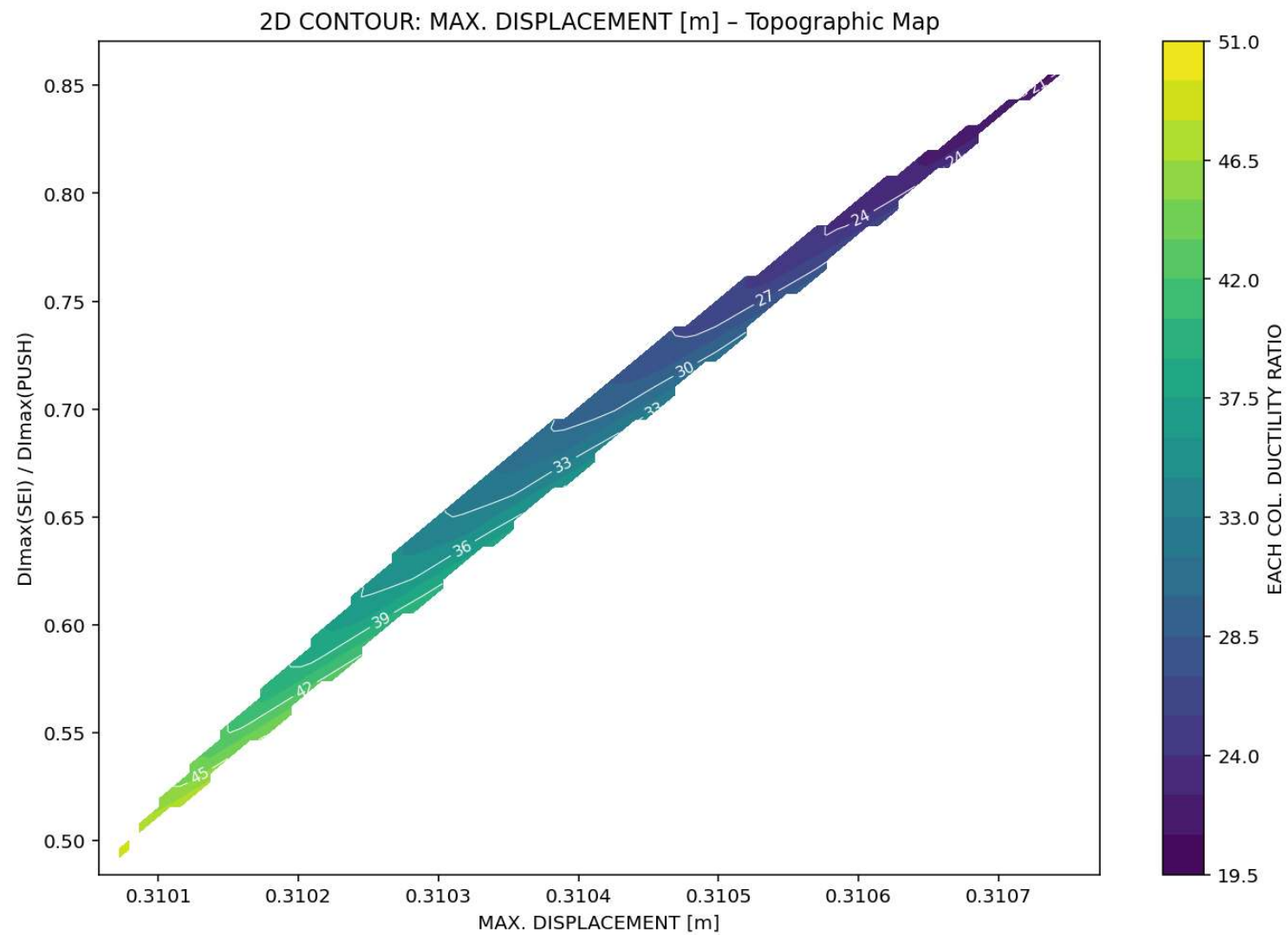
3D CONTOUR: $EDVR(SEI) / EDVR(CP)$ - $D_{max}(SEI) / D_{max}(CP)$ - EACH COL. DUCTILITY RATIO



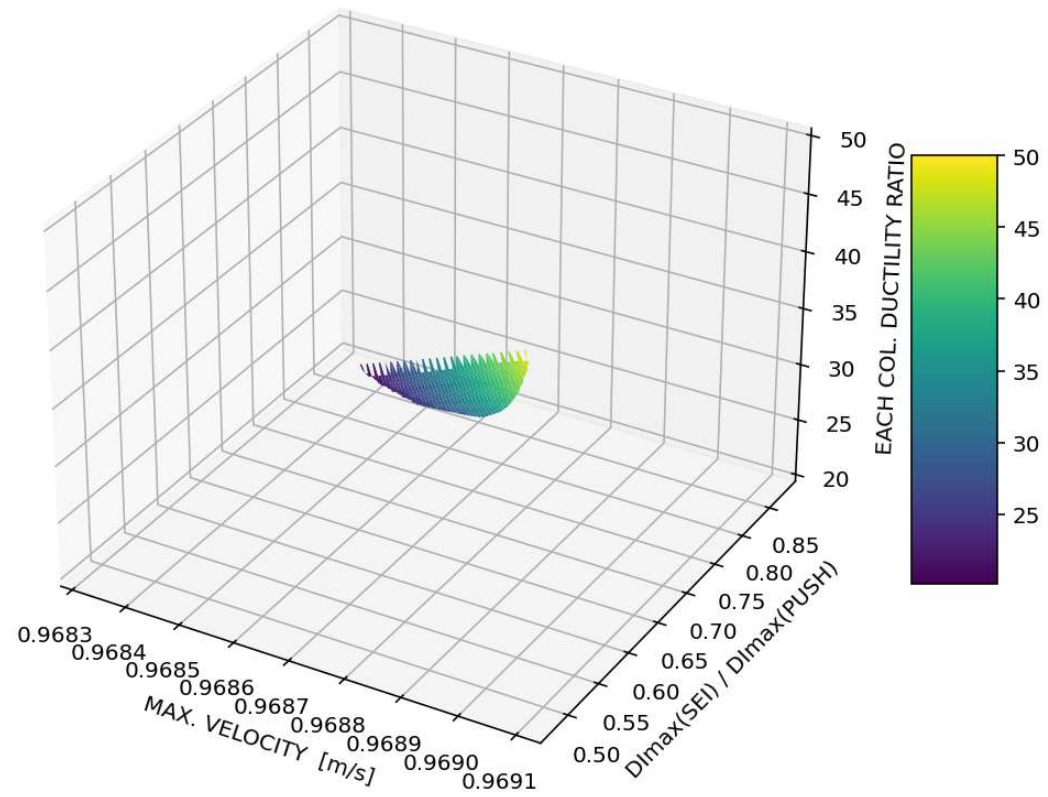


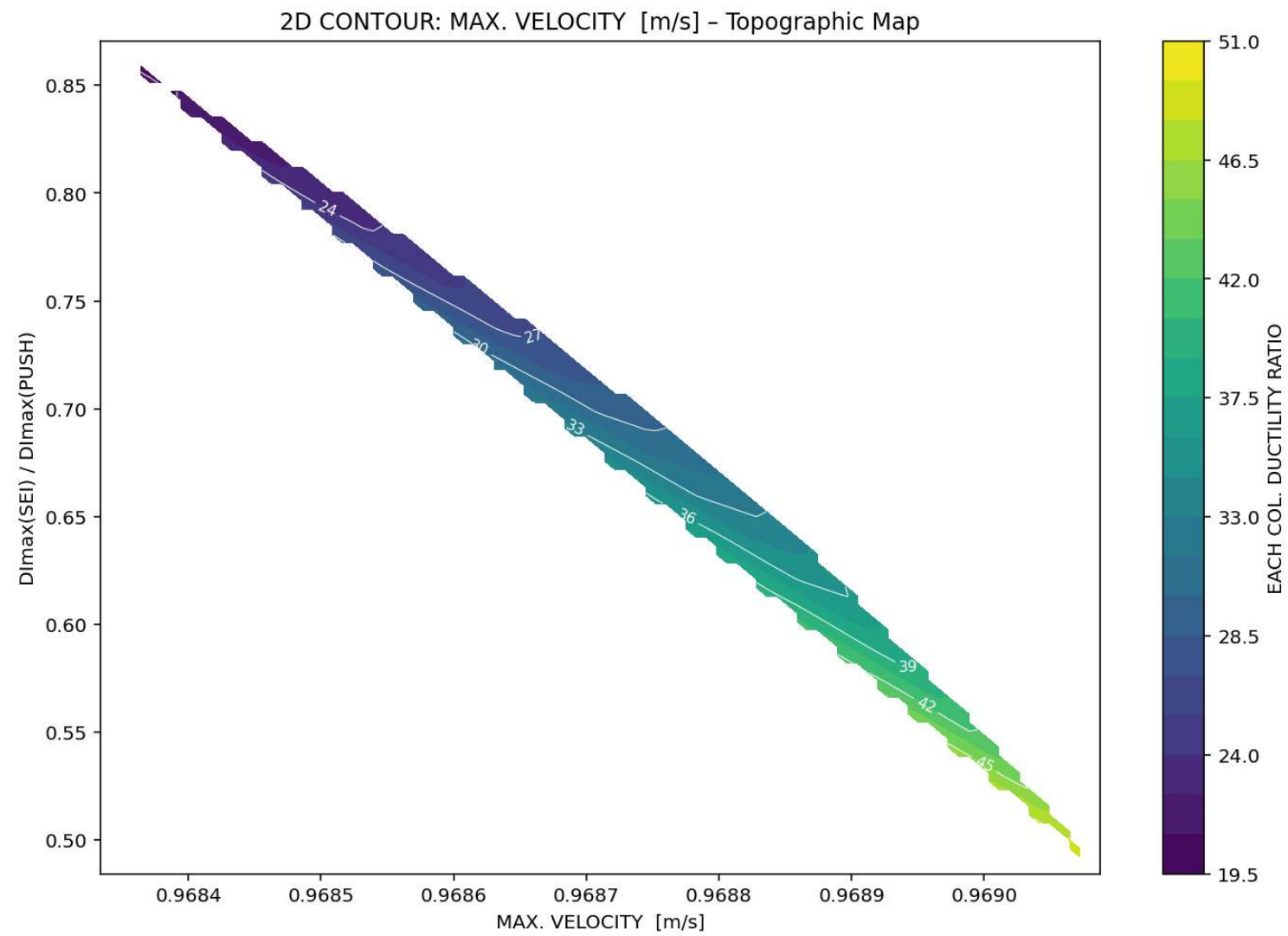
3D CONTOUR: MAX. DISPLACEMENT [m] - $D_{\max}(\text{SEI}) / D_{\max}(\text{PUSH})$ - EACH COL. DUCTILITY RATIO





3D CONTOUR: MAX. VELOCITY [m/s] - $D_{\max}(\text{SEI}) / D_{\max}(\text{PUSH})$ - EACH COL. DUCTILITY RATIO





3D CONTOUR: MAX. ACCELERATION [m/s²] - D_{lmax}(SEI) / D_{lmax}(PUSH) - EACH COL. DUCTILITY RATIO

